

Characteristic Geometry and Portfolio Choice

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Abstract

We study how characteristic geometry enters portfolio choice. In rolling U.S. equity estimates, nearly identical forecasts generate different portfolios because a fixed adjustment to factor variances imposes different exposure penalties across measurement conventions. Removing this adjustment nearly eliminates the weight gap. Within one fitted model, using the retained intercept in expected returns raises the gross Sharpe ratio from 0.39 to 2.51 over 594 common months; an instability adjustment raises it to 2.53. These findings connect exposure measurement to investment outcomes and document the value of using retained mean-return signals in portfolio construction.

Keywords: conditional factor models; firm characteristics; characteristic geometry; portfolio choice; covariance regularization.

JEL classification: G11, G12, C38, C58.

Disclosure Statement

Introduction

Firm characteristics are central to modern return prediction and portfolio management. Factor models and machine learning methods use large sets of characteristics to estimate expected returns. Their investment value, however, also depends on how those estimates enter portfolio construction. Different valuation ratios may favor many of the same stocks, so adding predictors need not diversify holdings. The investor must decide both which return forecasts to use and how to measure and control the associated portfolio exposures.

This paper studies the role of characteristic geometry in that decision. Our focus is how the scale and cross-sectional dependence of characteristics enter both the normalization of latent loadings and their separation from observed-factor loadings. Nearly identical return forecasts can lead to materially different portfolios when a fixed covariance adjustment is applied to different loading representations. A second experiment shows that a characteristic-dependent intercept already present in the fitted model can substantially improve gross portfolio performance when included in expected returns. These experiments separate the role of exposure penalties from the value of using additional fitted return information.

We build on instrumented principal components analysis (IPCA), which estimates latent return factors and models stocks' factor loadings as functions of firm characteristics (Kelly, Pruitt, and Su, 2019). Factor loadings measure a stock's exposure to the common return components. Our application also includes observed benchmark factors. The baseline makes the latent loading columns orthonormal and orthogonal to the observed-factor loading block under the Euclidean inner product on characteristic coefficients. We compare it with QZ-IPCA, which uses an estimated characteristic Gram matrix for both latent-block normalization and cross-block orthogonalization. Both models initially exclude the intercept; the intercept-inclusive specification is introduced below.

The Gram matrix averages characteristic cross-products and thus measures the scale and overlap of the stock-level exposures generated by the loadings. Using this characteristic metric makes the latent and observed-factor loading exposures orthogonal on average over the weighted estimation sample. The corresponding coefficient vectors can be orthogonal under the Euclidean metric without satisfying this exposure-based restriction. Cross-block orthogonalization removes from the observed-factor loadings their projection onto the latent span and reassigns that contribution to the latent factors. A matched adjustment preserves the complete fitted return, as shown in

Section I.D. The metric therefore governs attribution across blocks as well as the scale of the latent loadings. Factor-return covariances continue to enter portfolio risk separately.

A simple rescaling illustrates the portfolio mechanism. Doubling a factor’s return and halving its loadings leaves fitted stock returns unchanged. Transforming the forecasts and covariance consistently also preserves the implied stock-level expected returns and covariance. The same principle applies to matched cross-block transformations, provided the forecasts and joint factor covariance are transformed together. The portfolio routine, however, adds a fixed amount to each factor variance, a regularization step known as a factor-covariance ridge. Keeping this adjustment fixed across representations changes the quadratic penalty on portfolio factor exposures in the mean–variance objective. A change in loading scale or cross-block attribution can therefore change portfolio weights without changing the return forecast.

We examine this mechanism in rolling U.S. equity estimates with 132 characteristics. The characteristic metric is updated in each estimation window. The main comparison uses an exponentially weighted moving average (EWMA), which assigns greater weight to recent observations. The saved matrices show concentrated characteristic variation and changes in its distribution over time. Section III.A explains how preprocessing and averaging affect these measures. The characteristic count remains fixed throughout the comparison.

The EWMA comparison produces nearly identical forecasts but substantially different portfolio weights. Removing only the factor-covariance ridge, while retaining the forecasts and other risk inputs, reduces the maximum relative weight difference to approximately 0.01%. Reconstructing the covariance matrices from the saved factor histories verifies the inputs to this calculation. The results trace the weight differences to the interaction between the metric-dependent loading representation and covariance regularization. Comparing forecasts alone would miss this source of variation in portfolio allocations.

Realized performance provides a separate assessment of the investment strategies. Over June 1973–November 2024, the annualized gross Sharpe ratio is 0.64 for IPCA and ranges from 0.85 to 1.00 across the three reported QZ-IPCA metric specifications. For the EWMA specification used in the mechanism analysis, the Sharpe ratio is 0.91 and the paired 95% confidence interval for its difference from IPCA is $[-0.02, 0.58]$. These comparisons use the same months and returns before trading costs. Point estimates favor QZ-IPCA, but statistical significance depends on the metric

and resampling choices. The ridge-removal experiment explains the weight differences; assessing its performance consequences requires return paths for ridge-free or consistently transformed-ridge portfolios, which are not evaluated here.

The second experiment examines the investment value of the fitted intercept: the characteristic-dependent component of expected returns not attributed to factor means under the chosen decomposition. We retain the metric-based normalization and cross-block restrictions, make the intercept loading orthogonal to the latent loading block under the same metric, and impose a bound on its exposure. We call this specification No-Arbitrage Instrumented Principal Components Analysis (NA-IPCA). The name reflects the motivation for controlling departures from a zero-intercept factor-pricing benchmark. A positive budget limits, rather than eliminates, this component. Section I.C develops its economic interpretation and the additional conditions needed for a pricing interpretation.

Using the same NA-IPCA estimates, we compare three portfolio rules: one excludes the intercept from expected returns and its instability adjustment from risk; one restores the intercept to expected returns; and one also adjusts the risk matrix for variation in past intercept-loading estimates. Over 594 common months, their gross Sharpe ratios are 0.39, 2.51, and 2.53, respectively. Most of the performance difference comes from the expected-return input; the loading-instability adjustment contributes a smaller increment. This same-fit comparison documents the investment value of a fitted mean component that a factor-only allocation omits. Recent-period evidence is less precise; Section VI.D reports the uncertainty and sensitivity checks.

Our contribution is to connect characteristic-based return models to explicit choices in portfolio construction. Kelly, Pruitt, and Su (2019) use characteristics to estimate time-varying factor loadings; Lettau and Pelger (2020) incorporate expected-return information into factor estimation; and Gu, Kelly, and Xiu (2020) study return prediction and its investment value using machine learning. We complement these approaches by isolating how the metric-based loading decomposition and the use of fitted mean components affect allocations and realized returns. The evidence separates differences in fitted models from differences in their use, allowing investors to evaluate the particular portfolio decision responsible for a change in holdings or performance.

The analysis also contributes to research on estimation risk, covariance shrinkage, and portfolio constraints (Jagannathan and Ma, 2003; DeMiguel, Garlappi, and Uppal, 2009; Kan and Zhou,

2007; Ledoit and Wolf, 2017), and complements the direct use of characteristics in portfolio policies (Brandt, Santa-Clara, and Valkanov, 2009). We show how to preserve a chosen stock-level risk adjustment when the factor representation changes: transform the factor-level adjustment with the covariance, or specify the adjustment directly in stock space. This separates a change in factor representation from a deliberate change in portfolio exposure penalties. The companion working paper Liu (2026) develops the identification and inference framework. The present paper states the representation result needed for the portfolio analysis and evaluates the empirical comparisons using paired strategy returns.

The remainder of the paper introduces the models, estimation design, and portfolio rules; compares characteristic geometry, forecasts, and holdings; derives the regularization mechanism; and evaluates the intercept extension and robustness of the performance results. The Internet Appendix contains derivations, additional tests, and replication details.

I. Models, Characteristic Metrics, and Portfolio Rules

A. The return model and the attribution rule

We first specify what the model estimates, then distinguish its normalization from the rule used to turn forecasts into portfolios. At formation date t , N_t is the number of eligible stocks and L the number of characteristics. The matrix $Z_t \in \mathbb{R}^{N_t \times L}$ has row i equal to $z'_{i,t}$. The estimation window is $\mathcal{W}_t = \{t - 120, \dots, t - 1\}$ in monthly indices. For stock $i = 1, \dots, N_s$ and training date $s \in \mathcal{W}_t$, next-month excess returns satisfy

$$r_{i,s+1} = z'_{i,s} \Gamma_\beta f_{s+1} + z'_{i,s} \Gamma_\delta g_{s+1} + z'_{i,s} \Gamma_\alpha + \varepsilon_{i,s+1}, \quad (1)$$

where $f_{s+1} \in \mathbb{R}^K$ contains the latent factors and $g_{s+1} \in \mathbb{R}^{K_o}$ contains the observed factors, with K and K_o denoting their number. The disturbance $\varepsilon_{i,s+1}$ is stock-specific. The loadings $\Gamma_\beta \in \mathbb{R}^{L \times K}$, $\Gamma_\delta \in \mathbb{R}^{L \times K_o}$, and $\Gamma_\alpha \in \mathbb{R}^L$ are constant within a window but re-estimated across windows. In the empirical application, the observed block comprises the five Fama–French factors ($K_o = 5$; Fama and French, 2015), and we use $K = 3$ and $L = 132$. Window indices on the loadings and on the training-standardized characteristics are suppressed here. The latent block describes common

variation not assigned to the observed block under the declared normalization. The intercept allows an additional characteristic-dependent component in the fitted mean specification. Realized factor variation in (1) must be forecast before forming a portfolio.

Write $h_{s+1} = \Gamma_\beta f_{s+1} + \Gamma_\delta g_{s+1} + \Gamma_\alpha$ for the complete characteristic index. Its decomposition is not unique. An invertible transformation of the latent loading basis, offset by a transformation of the factors, leaves h_{s+1} unchanged. Overlap with observed-factor and intercept loadings can likewise be transferred to the latent factors. A normalization specifies which representative is reported.

For a positive-definite metric $Q \in \mathbb{R}^{L \times L}$, we impose

$$\Gamma'_\beta Q \Gamma_\beta = I_K, \quad \Gamma'_\beta Q [\Gamma_\delta, \Gamma_\alpha] = 0. \quad (2)$$

These restrictions normalize the latent block and separate it from the retained blocks in characteristic space. They leave rotations within the latent block unresolved; the object of interest is the loading class modulo those rotations. They do not impose orthogonality between the observed block and the intercept.

We next explain how the metric measures exposure, then use that measurement to define the intercept-budget specification and its pricing motivation.

B. What the characteristic metric measures

Let $r_{s+1} = (r_{1,s+1}, \dots, r_{N_s,s+1})'$ stack the stock returns. Define the monthly Gram matrix and managed-return moment by

$$W_s = \frac{Z'_s Z_s}{N_s}, \quad X_s = \frac{Z'_s r_{s+1}}{N_s}. \quad (3)$$

For a coefficient direction $v \in \mathbb{R}^L$, $v' W_s v = N_s^{-1} \sum_i (z'_{i,s} v)^2$ measures average squared exposure. It is a second moment, and equals a variance only after the relevant exposure is centered. Correlated characteristics can therefore give two equally long coefficient vectors very different exposure magnitudes. Marginal rank transformations do not remove this cross-characteristic overlap.

The same calculation explains what orthogonality means. For loading directions $u, v \in \mathbb{R}^L$, the cross-sectional product of their exposures is $u' W_s v = N_s^{-1} \sum_i (z'_{i,s} u)(z'_{i,s} v)$. Thus $u' v = 0$ need not make their stock exposures orthogonal. Averaging these products gives the corresponding

interpretation of $u'\widehat{Q}_t v$. The metric aligns loading comparisons with measured characteristic exposure. Portfolio risk additionally depends on factor-return covariances: orthogonal loading directions need not generate uncorrelated portfolio payoffs.

QZ-IPCA uses an estimated trailing average $\widehat{Q}_t$ of the W_s in (2); the Euclidean comparator uses I_L . The empirical rules are a 120-month exponentially weighted average, a 120-month equal-weight average, and a 60-month equal-weight average, each calculated within the rolling estimation window. The notation Q_Z refers to a population characteristic metric, whereas $\widehat{Q}_t$ is the information-date-specific object used in the rolling experiment. This distinction keeps a population identification rule separate from its estimated implementation.

Updating $\widehat{Q}_t$ changes this exposure convention using information already in the model's trailing estimation sample. The window length, averaging weights, and preprocessing jointly determine what the updated ruler measures. Section III.A documents its evolution in the production panel. Section IV then shows exactly when such a convention changes the portfolio penalty and when consistent transport preserves the investment rule.

C. The pricing motivation for No-Arbitrage IPCA

No-Arbitrage Instrumented Principal Components Analysis (NA-IPCA) combines a metric-indexed loading decomposition with an explicit budget for the retained intercept. Its pricing motivation is to control the characteristic-dependent expected return left outside the factor-mean components. IPCA already accommodates a characteristic-dependent intercept (Kelly, Pruitt, and Su, 2019); the specification here adds a declared metric for separating the loading blocks and a bound on residual mean exposure.

To establish this interpretation, let $\mathcal{I}_s$ denote investors' information at date s , and let $z_{i,s} \in \mathbb{R}^L$ collect the L observed characteristics of stock i . The vectors f_{s+1} and g_{s+1} contain the latent and observed factors, respectively, while Γ_β , Γ_δ , and Γ_α are the characteristic-loading coefficients for the latent, observed-factor, and intercept blocks. For this population calculation, take the loading coefficients to be fixed, the characteristics to be $\mathcal{I}_s$ -measurable, and the idiosyncratic disturbance in (1) to have zero conditional mean. Define the conditional factor means by $m_{f,s} := \mathbb{E}(f_{s+1} \mid \mathcal{I}_s)$

and $m_{g,s} := \mathbb{E}(g_{s+1} \mid \mathcal{I}_s)$. Taking conditional expectations of the return equation gives

$$\mathbb{E}(r_{i,s+1} \mid \mathcal{I}_s) = z'_{i,s} \Gamma_\beta m_{f,s} + z'_{i,s} \Gamma_\delta m_{g,s} + \alpha_{i,s}, \quad \alpha_{i,s} := z'_{i,s} \Gamma_\alpha, \quad (4)$$

where $r_{i,s+1}$ is the stock's excess return and $\alpha_{i,s}$ is its retained conditional-mean component. The zero-intercept benchmark attributes the entire conditional mean to the factor terms; a nonzero intercept records what remains under the chosen decomposition.

This interpretation connects the intercept to conditional asset pricing. Under an appropriate factor-pricing specification, residual conditional means are pricing errors. In Zaffaroni (2025), asymptotic no-arbitrage and additional regularity conditions imply square-summability of cross-sectional pricing errors, rather than requiring every intercept to vanish. The analysis also examines omitted factors and misspecified conditional dynamics, highlighting the challenges they pose for interpreting empirical alpha. For our application, this distinguishes a retained mean component from a pricing error relative to an economically specified factor benchmark.¹

NA-IPCA makes control of the retained mean operational through the intercept-exposure budget

$$\Gamma'_\alpha Q \Gamma_\alpha \leq \delta_\alpha, \quad (5)$$

where $Q \succ 0$ is the declared characteristic-space metric and $\delta_\alpha \geq 0$ is the permitted exposure budget. When Q is a trailing weighted average of characteristic Gram matrices, the quadratic form equals the corresponding training-weighted average of squared stock-level intercepts. A zero budget removes the intercept block; a positive budget allows residual mean exposure while limiting its magnitude. The finite-sample budget and the cross-sectional summability condition both control residual means, but impose different restrictions. When the factor-pricing benchmark identifies the retained intercept with conditional pricing errors, the budget provides an explicit measure of the permitted departure from that benchmark.

The empirical value is an explicit link between model decomposition and portfolio design.

¹A standard pricing benchmark uses a strictly positive stochastic discount factor (SDF), M_{s+1} , which converts future payoffs into date- s values. It prices excess returns through $\mathbb{E}(M_{s+1} r_{i,s+1} \mid \mathcal{I}_s) = 0$; see Cochrane (2001), Chapters 1 and 4. Connecting the factor-mean terms in (4) to this benchmark requires restrictions on the factor span and the normalization linking factor means to risk premia.

Internet Appendix IA.4.1 translates the budget into a bound on a portfolio’s intercept contribution. Section V then evaluates whether including the estimated intercept in the portfolio mean improves realized performance, separately from the effect of a loading-instability adjustment. This comparison identifies the investment contribution of the retained mean; interpreting it as mispricing requires a specified pricing benchmark. Table I distinguishes the fitted models from these alternative portfolio uses of one NA-IPCA fit.

Table I. Model setup and the comparison each specification permits. The conditional family has three latent and five observed factors. All loadings are re-estimated in each 120-month window. The metric enters the normalization in equation (2); its empirical constructions are defined in Section II.B. NA-IPCA denotes No-Arbitrage IPCA, the metric-indexed intercept-budget specification. The portfolio input comparison uses one NA-IPCA fit, not three models; IPCA here denotes the zero-intercept comparator.

Specification	Loading restrictions	Empirical question
IPCA	$Q = I_L, \Gamma_\alpha = 0$	Euclidean reference for normalization and portfolios.
QZ-IPCA	$Q = \widehat{Q}_t, \Gamma_\alpha = 0$	Does a metric change alter the portfolio rule when forecasts remain nearly fixed?
NA-IPCA	$Q = \widehat{Q}_t, \Gamma'_\alpha Q \Gamma_\alpha \leq 0.035$	What changes when the portfolio retains the fitted intercept in its mean and adjusts for its instability?

D. Matched representations and fitted returns

The following result states exactly what is held fixed in a change of metric. It specializes the construction of a normalized representative in Liu (2026).

PROPOSITION 1 (Preservation of the complete fitted index): Suppose Γ_β has rank K and $Q \succ 0$. Set $C = \Gamma'_\beta Q \Gamma_\beta$, $D = C^{-1} \Gamma'_\beta Q \Gamma_\delta$, and $a = C^{-1} \Gamma'_\beta Q \Gamma_\alpha$, of dimensions $K \times K$, $K \times 5$, and K , respectively. Let $C^{1/2}$ be the symmetric positive-definite square root. The transformed blocks

$$\begin{aligned} \Gamma_\beta^Q &= \Gamma_\beta C^{-1/2}, & \Gamma_\delta^Q &= \Gamma_\delta - \Gamma_\beta D, & \Gamma_\alpha^Q &= \Gamma_\alpha - \Gamma_\beta a, \\ f_{s+1}^Q &= C^{1/2}(f_{s+1} + D g_{s+1} + a) \end{aligned} \tag{6}$$

satisfy (2) and preserve h_{s+1} for every date. Consequently, fitted stock returns and residuals are unchanged.

The four transformations have distinct roles. First, $\Gamma_\beta^Q = \Gamma_\beta C^{-1/2}$ expresses the same latent span in a Q -orthonormal basis. It standardizes the loading directions in the declared exposure geometry, with the corresponding change in factor units absorbed by $C^{1/2}$. Second, $\Gamma_\delta^Q = \Gamma_\delta - \Gamma_\beta D$ removes from the observed-factor loading block its Q -projection onto the latent span. The observed factors g_{s+1} retain their labels and realizations, while the overlapping loading contribution $\Gamma_\beta D g_{s+1}$ is reassigned to the latent block. The transformation therefore changes attribution without discarding the return variation associated with those observed factors. Third, $\Gamma_\alpha^Q = \Gamma_\alpha - \Gamma_\beta a$ makes the same adjustment to the intercept. The component $\Gamma_\beta a$ is absorbed into the level of the latent factors, leaving an intercept loading that is Q -orthogonal to the latent span. The retained intercept thus measures the characteristic-dependent mean component left outside that span under the declared decomposition. Finally, $f_{s+1}^Q = C^{1/2}(f_{s+1} + D g_{s+1} + a)$ compensates for all three loading adjustments. The term $D g_{s+1}$ carries the transferred observed-factor contribution, a carries the transferred intercept, and $C^{1/2}$ offsets the normalization of the latent loadings. Consequently, latent factor realizations and means depend on the chosen representation even though the complete fitted index does not. The normalization follows from $C^{-1/2} C C^{-1/2} = I_K$ and the definitions of D and a . Substituting (6) into the fitted index adds $\Gamma_\beta D g_{s+1} + \Gamma_\beta a$ to the latent contribution and subtracts the same terms from the retained blocks. The terms cancel. Thus, the transformation changes the allocation across blocks, not the complete fitted return.

This gives a benchmark for the empirical comparison. For matched representations, fitted-return agreement is an algebraic identity; agreement of complete return forecasts also follows when factor forecasts are transformed by the same affine rule. For separately initialized fits, whether the

estimates recover the same fitted-index class must be checked in the data. The diagnostics below establish this comparison before examining differences in portfolio construction.

The distinction clarifies the role of replacing I_L by the characteristic metric Q_Z , implemented through $\widehat{Q}_t$. The Euclidean IPCA comparator separates loading blocks by the inner product of their coefficient vectors. The characteristic metric instead measures separation through the stock-level exposures those coefficients generate, as shown in Section I.B. Directions that are orthogonal under I_L can therefore overlap in measured exposure. Changing the metric changes which parts of the observed-factor and intercept loadings are retained outside the latent span, even when the complete fitted index is unchanged. The economic object being measured is exposure across stocks, rather than coefficient length in the recorded characteristic coordinates.

Our empirical contribution is to trace this distinction from loading decomposition to portfolio inputs and investment outcomes. A portfolio rule built entirely from invariant inputs remains unchanged under a matched transformation. A rule that uses selected mean components, loading-instability measures, or a penalty held fixed in factor coordinates can respond to the metric. Comparing these rules against the same complete fitted-index benchmark allows us to examine where measured characteristic geometry enters the investment decision. Section IV develops this comparison, separating the information reproduced by the fitted model from the allocation and regularization choices used to construct portfolios.

The intercept budget in (5) gives the metric an additional role. Under I_L , the budget bounds the squared coefficient length of the retained intercept; under a Gram-based metric, it bounds the corresponding average squared stock-level intercept exposure. The same numerical budget therefore need not represent the same exposure restriction across metrics. Unlike a change of representation alone, the budget can exclude fitted-index classes. Strict slackness preserves feasibility under sufficiently small matched metric changes; near a binding budget, changing the metric can also change the admissible fitted means.

E. Estimation and the scope of the companion theory

Within each window $\mathcal{W}_t$, the ideal estimator minimizes stock-level squared errors subject to metric normalization, cross-block orthogonality, and the intercept-exposure budget. Up to a

parameter-independent constant, the criterion can be written in managed-moment form as

$$\sum_{s \in \mathcal{W}_t} N_s (X_s - W_s h_{s+1})' W_s^+ (X_s - W_s h_{s+1}), \quad (7)$$

where W_s^+ denotes the Moore–Penrose inverse. Expanding the quadratic forms establishes the equivalence even when W_s is singular, because X_s lies in its range. Managed moments therefore compress the data without changing the least-squares criterion. The weight N_s gives equal weight to stock-month observations. These criterion weights serve a different purpose from the metric Q , which determines the normalization and separation of the loading blocks.

The moment-ridge adjustment used in implementation adds a penalty on the complete fitted index. Because this penalty depends only on h_{s+1} , it preserves the criterion’s invariance under matched changes of representation. Internet Appendix IA.1 derives the adjustment and documents its tuning parameter and the separate numerical stabilizations used in the block solves.

Estimation combines alternating factor and loading updates with metric normalization, cross-block orthogonalization, and enforcement of the intercept budget when needed. The rolling NA-IPCA implementation uses the previous window’s estimates as starting values. A separate common-initialization exercise examines whether differences across fits persist when starting values are held fixed. Internet Appendix IA.1.3 reports the stopping diagnostic, and Table IA.I reports the initialization comparison. The covariance ridges introduced below enter portfolio construction, rather than the estimation criterion.

The companion paper (Liu, 2026) provides a foundation for interpreting the estimated loading decomposition. It establishes local identification of loading classes modulo latent rotations and develops inference for an exactly profiled, fixed-weight estimator that accounts for estimation of the characteristic metric. Under covariance alignment and the accompanying regularity conditions, the population criterion target coincides with the structural loading class. These results support inference on features of the estimated decomposition. The empirical analysis examines the investment rules constructed from that decomposition, evaluating their performance using paired monthly strategy returns and the time-series resampling procedure described below.

F. Portfolio construction

We translate each model’s estimated expected returns and covariances into stock positions using a common mean–variance rule. The construction connects the model’s exposure measurements to an investment decision: which positions to hold, and how much estimated risk to accept for their forecast returns. We first describe the inputs and regularization, then explain how the characteristic metric enters the resulting allocation.

Write $F_{s+1} = (f'_{s+1}, g'_{s+1})' \in \mathbb{R}^{K+K_o}$ for the combined vector of latent and observed factors. At formation date t , let $B_t = [Z_t \widehat{\Gamma}_{\beta_t}, Z_t \widehat{\Gamma}_{\delta_t}] \in \mathbb{R}^{N_t \times (K+K_o)}$ collect their exposures across the N_t eligible stocks. The factor-return forecast $\widehat{m}_{F,t}$ is the arithmetic mean of the factor histories in the 120-month training window. Their centered sample covariance, $\widehat{C}_{F,t}$, uses denominator 120 – 1. Both moments combine estimated latent factors and observed factor realizations, retaining all latent–observed cross-covariances. Loading orthogonality does not imply uncorrelated factor returns. The window length and equal-weight moment calculations are common across metric rules; the fitted factor histories are model-specific.

Estimated covariances can assign very little risk to combinations of positions whose apparent safety reflects sampling error. An inverse-covariance allocation can amplify that error into large, unstable positions, even when the covariance estimate is nonsingular. Covariance shrinkage and portfolio constraints are established responses to this estimation problem (Jagannathan and Ma, 2003; Ledoit and Wolf, 2017). We use a simple diagonal ridge: adding $\lambda_F > 0$ to each factor-covariance eigenvalue improves conditioning. A separate stock-space ridge $\lambda_\Sigma > 0$ regularizes the allocation across individual stocks. This familiar stabilization device provides the setting in which we examine how loading units affect portfolio choice.

Let $\widehat{D}_t$ be the diagonal matrix of estimated stock residual variances. The baseline uses $\lambda_F = \lambda_\Sigma = 0.01$ across models, metric rules, and formation dates, without model-specific tuning. Define the stock-return forecast $\widehat{\mu}_t$, regularized factor covariance $\Sigma_{F,t}$, stock-level risk matrix A_t , and raw portfolio weights w_t by

$$\widehat{\mu}_t = B_t \widehat{m}_{F,t}, \quad \Sigma_{F,t} = \widehat{C}_{F,t} + \lambda_F I_{K+K_o}, \quad (8)$$

$$A_t = B_t \Sigma_{F,t} B_t' + \widehat{D}_t + \lambda_\Sigma I_{N_t}, \quad w_t = A_t^{-1} \widehat{\mu}_t. \quad (9)$$

The exposure matrix maps factor means into expected stock returns and factor covariances into stock-level covariation. The residual variances and stock-space ridge complete the risk matrix used in the allocation.

The raw weights maximize $v'\widehat{\mu}_t - \frac{1}{2}v'A_tv$ over stock-position vectors $v \in \mathbb{R}^{N_t}$. They therefore balance forecast returns against the risk of the entire portfolio, including covariation across stocks. For $w_t \neq 0$, we implement $w_t/\|w_t\|_1$, setting total absolute stock exposure to one. This unit-gross normalization makes position scale comparable across strategies while preserving the relative positions from the unconstrained solve.

Expanding the quadratic risk term makes the two regularization channels explicit:

$$\begin{aligned} v'A_tv &= (B'_t v)' \widehat{C}_{F,t} (B'_t v) + v' \widehat{D}_t v \\ &\quad + \lambda_F \|B'_t v\|_2^2 + \lambda_\Sigma \|v\|_2^2. \end{aligned}$$

The factor ridge penalizes aggregate factor exposures $B'_t v$ in their reported units; the stock-space ridge penalizes the squared magnitude of stock positions. Since the other terms are nonnegative, $v'A_tv \geq \lambda_\Sigma \|v\|_2^2$, so A_t is positive definite and the raw solve is unique. Regularization thus specifies exposure penalties as well as stabilizing the numerical calculation.

The characteristic metric determines the units and cross-block separation underlying these exposures. Our Euclidean IPCA comparator normalizes latent loadings and separates them from the retained blocks using coefficient-space inner products. QZ-IPCA instead uses the trailing Gram metric in (2). Its latent directions have unit training-weighted average squared stock exposure and are orthogonal to the retained blocks under the same exposure measure. These are averages over the metric's training observations, not date-by-date restrictions. The practical rationale for the Gram rule is to account for characteristic scale and overlap when measuring loading directions, while factor-return covariances continue to determine their contribution to portfolio risk.

This distinction has a direct consequence for the ridge. When the intercept is excluded from both IPCA and QZ-IPCA, a matched change of representation with consistently transformed factor moments preserves the stock-return forecast and the unregularized stock covariance $B_t \widehat{C}_{F,t} B'_t$. The factor ridge nevertheless contributes $\lambda_F B_t B'_t$ in stock space, which can change under the same transformation. Equal numerical ridge values therefore need not impose equal stock-level exposure penalties. Section IV derives this channel and shows how transporting the adjustment with the

factor coordinates preserves the penalty. It provides a way to separate new forecasting information from a change in the investment rule.

The empirical comparisons follow this distinction. Section III evaluates forecast agreement and realized portfolio performance across metric rules, and Section IV examines the source of weight differences. NA-IPCA adds the intercept budget to the metric-based decomposition. Its baseline portfolio mean here includes only the factor terms; Section V holds one NA-IPCA fit fixed and separately adds the retained intercept to expected returns and a loading-instability adjustment to risk. These exercises assess exposure measurement, retained mean information, and risk adjustment without conflating their investment contributions.

The benchmark models vary the source of factor information within the same allocation protocol. FF5 uses the observed market, size, value, profitability, and investment factors, with characteristic-instrumented stock exposures $B_t = Z_t \widehat{\Gamma}_{\delta t}$. It is the observed-factor-only reference, rather than a separate stock-by-stock beta regression. Risk-premium PCA (RP-PCA; Lettau and Pelger, 2020) extracts latent factors from characteristic-managed returns using both covariance and mean information, with $B_t = Z_t \widehat{\Gamma}_{\beta t}$. It complements covariance-based PCA by allowing low-variance directions to matter for expected returns. The archived centered-PCA configuration produces a zero factor-mean forecast and hence no nonzero unit-gross allocation under this rule; Internet Appendix IA.2.2 explains its exclusion from the reported portfolio comparison.

The latent-only specifications use eight factors, matching the three latent and five observed factors in the conditional models; FF5 retains five. Each implemented strategy uses its own factor histories, exposures, and residual variances, with common moment conventions, numerical ridge values, and unit-gross normalization. Internet Appendix IA.2 gives the benchmark estimators. All reported strategy returns are returns on stock portfolios, not on the factor series themselves.

II. Data and Empirical Design

A. Panel, estimation windows, and model comparisons

We study monthly U.S. stock returns using the characteristic data of Jensen, Kelly, and Pedersen (2023), which provide a standardized collection of stock-level return predictors from the finance literature. Our sample contains $L = 132$ characteristics and combines stocks in the mega-, large-,

and small-capitalization categories. Micro-capitalization stocks are excluded. This common panel provides the characteristic information used to estimate the loading models and construct their associated portfolios.

We exclude stock-month observations for which more than 30% of the characteristic values are missing. Each month, we rank the available observations for each characteristic across stocks and map the ranks to the interval $[-0.5, 0.5]$. This transformation places characteristics measured in different units on a common scale and limits the influence of extreme observations. Any remaining missing characteristic inputs are assigned zero before window-specific standardization. Within each estimation window, we further center and scale the characteristics using training-sample means and standard deviations, and apply the same transformation at portfolio formation. The preprocessing rule is common across models, holding characteristic scaling fixed when we compare alternative loading metrics.

Models are re-estimated in 120-month rolling windows and evaluated on next-month excess returns. The out-of-sample panel spans 618 months and contains 1,244,002 stock-month observations from 12,703 distinct stocks. The median monthly cross section contains 1,990 stocks, with monthly counts ranging from 1,335 to 3,150. Let N_t denote the number of eligible stocks at date t . The managed-moment formulation accommodates this unbalanced panel, allowing the cross section to change over time without requiring stocks to be present throughout an estimation window.

The conditional-model comparisons vary the loading geometry and the treatment of the intercept. Throughout the empirical analysis, IPCA denotes the Euclidean specification with $K = 3$ latent factors and $K_o = 5$ observed factors. QZ-IPCA retains these factor blocks and uses the characteristic Gram metric for loading normalization and cross-block orthogonality. NA-IPCA additionally includes a characteristic-dependent intercept subject to $\delta_\alpha = 0.035$. The budget is slack at the estimated solution in every window, so the empirical mechanism analysis concerns decomposition and portfolio construction in the slack-budget regime.

The remaining benchmarks provide comparisons across factor specifications. FF5 uses five observed factors with stock exposures restricted to a common characteristic-loading map. It is a pooled characteristic-by-factor regression, not a two-pass risk-premium regression. PCA and RP-PCA instead extract eight latent factors from the characteristic-managed return panel, without an observed block. We implement the RP-PCA estimator of Lettau and Pelger (2020), setting

the pricing-weight parameter to $\gamma_{RP} = 3$ throughout the rolling experiment. This positive weight gives mean-return information additional influence in factor extraction, allowing directions with modest covariance variation to enter the estimated span. The conditional specifications and the PCA benchmarks therefore share a total of eight factors, while differing in whether observed factor labels enter estimation. All models use the same rolling-window protocol and the portfolio construction rule described above. Internet Appendix IA.2 specifies the benchmark estimators and the factor histories passed to that rule. Finally, the intercept-off exercise holds the NA-IPCA fit fixed and changes its use in portfolio construction. It therefore isolates a portfolio-input choice within the fitted model, rather than introducing another estimated model.

B. Samples and metric constructions

Each 120-month model fit produces one intercept-loading estimate, $\widehat{\Gamma}_{\alpha t}$. The loading-instability adjustment uses up to 24 preceding monthly estimates, excluding the current one. Its observations are loading vectors from successive fits, not the monthly factor returns within a single fit.

The rolling series of loading estimates begins in June 1973. The available-history sample retains all 618 formation months through November 2024, using the available prior loading estimates while the 24-month history accumulates.² The strict-24 sample begins in June 1975, the first formation month with 24 prior estimates. It contains 594 formation months through November 2024 and ensures that every instability matrix uses the same number of historical estimates. The return model continues to use a 120-month training window in both samples. Within each comparison, all strategies are evaluated on the same formation dates; results for the two samples are reported separately.

Dates throughout refer to portfolio formation, with payoffs realized in the following month. At formation t , the last training characteristics Z_{t-1} are paired with returns and observed factors realized at t ; current characteristics Z_t generate exposures for the $t + 1$ forecast. Figure IA.1 in Internet Appendix IA.3 displays the two input histories and their formation/payoff dates, including the first strict-24 observation.

²With two to 23 prior loading estimates, the adjustment uses their sample covariance. With fewer than two, this covariance cannot be computed, so only the additional instability-risk term is set to zero. The intercept remains in the mean forecast, and the baseline portfolio risk components are retained.

For historical context, gray bands in the time-series figures mark recessions dated by the National Bureau of Economic Research (NBER), from the month following a business-cycle peak through the trough month. The bands follow the displayed formation dates and serve as retrospective annotations, not inputs to the investment rules.

We compare three characteristic metrics, each constructed within the 120-month estimation window. The EWMA metric averages the training-month Gram matrices with a 60-month half-life, using the weights in (IA18). The equal-weight alternatives average either all 120 months or the final 60 months; the latter is the rolling-mean metric.³ All three constructions use the moment-ridge-adjusted Gram matrices, without additional trace normalization.

The forecast-agreement and ridge-removal diagnostics use EWMA, as does the intercept-inclusive portfolio. This common metric links the mechanism analysis and the intercept-input comparisons under the same exposure convention. The baseline paired performance comparison evaluates rolling-mean QZ-IPCA relative to IPCA. We apply the same paired bootstrap procedure to the other two metric constructions to assess dependence on the metric rule. Table II identifies the metric and sample for each exercise. The additional metric comparisons and within-fit input exercises are retrospective analyses of previously estimated models. EWMA’s role follows the construction used in the mechanism diagnostics, rather than its realized performance; all three metric rules are retained without selection by statistical significance.

C. A common portfolio policy

Each raw inverse-covariance solve is converted to unit gross exposure:

$$G_t = \sum_i |w_{i,t}|, \quad w_t^u = \frac{w_t}{G_t}, \quad r_{t+1}^u = (w_t^u)' r_{t+1}. \quad (10)$$

We require $G_t > 0$. This is a time-varying leverage policy, not merely a change in reporting units. It puts all strategies on a common gross budget without forcing equal volatility. Let $\bar{r}^u$ and $s(r^u)$ denote the sample mean and sample standard deviation of monthly unit-gross returns over the

³The replication files label the 120-month equal-weight construction “expanding.” This label denotes a fixed-length rolling average, not an expanding-history estimator.

Table II. From research question to evidence. All comparisons use Core stocks and unit-gross portfolios unless otherwise stated. The available-history sample has 618 formation months (June 1973–November 2024); the strict-24 sample has 594 (June 1975–November 2024). Metric rules and portfolio inputs are changed in separate exercises.

Question	Comparison and what is held fixed	Results
How dispersed and stable are the measured exposures?	All 618 saved EWMA Gram matrices; the feature count remains 132.	Figure 1; Table IA.III
Why can nearly equal forecasts give different weights?	EWMA QZ-IPCA versus IPCA; 618 windows. Remove only the factor ridge from saved portfolio inputs.	Section III.B; Table V; Figure 2
Does a different metric improve realized performance?	Each of three QZ-IPCA rules versus IPCA, on the same months; 618 and 594 months reported separately.	Tables III, IV, and IA.II
Which intercept input carries the gain?	One EWMA NA-IPCA fit; omit both inputs, restore the mean, then add instability risk; 594 months.	Tables VII and VI
How do full strategies compare?	Separately fitted strategies on 594 common months; all three QZ-IPCA rules shown.	Table VIII; Figure 3
Do forecast rankings explain portfolio rankings?	Stock-level forecast metrics on the 594-month common sample; QZ-IPCA uses rolling 60.	Table IA.IV; Section VI.B
How sensitive are the conclusions?	Common initialization; 12/24/60-month resampling blocks; instability multiplier and turnover-proxy checks.	Tables IA.I, IA.VI, and IA.V; Figure IA.2

stated evaluation sample. Annualized mean and volatility are $12\bar{r}^u$ and $\sqrt{12}s(r^u)$; the Sharpe ratio is $\sqrt{12}\bar{r}^u/s(r^u)$. Maximum drawdown is calculated from the compounded return path.

The label IAB (Integrated Alpha–Beta) denotes the complete intercept-inclusive portfolio. Its large raw exposure makes the leverage policy consequential: Internet Appendix IA.7.4 reports raw and normalized results separately. The reported strategy uses (10), not the unnormalized inverse-covariance solve. An archived PCA configuration produces near-zero expected-return inputs and gross exposure below 4×10^{-16} , so its normalized weights are numerically unstable. We exclude that configuration from portfolio rankings rather than interpret it as evidence against PCA.

The saved cost calculation deducts a proportional charge on raw turnover divided by current gross exposure. That object is not generally the turnover of w_t^u : actual trading also depends on prior normalized holdings and their return-induced drift. We consequently emphasize gross returns. Internet Appendix IA.4.4 preserves the cost calculation as a sensitivity proxy, not as a verified implementable net-return estimate. Stock-level holdings, financing, borrowing fees, and capacity are needed to complete the economic evaluation.

D. Inference and replication

We compare strategies on the same months and resample paired return records in moving blocks. The archived tests use 12-month blocks and 10,000 replications, preserving within-block time dependence and cross-strategy dependence. Percentile intervals describe the statistic’s resampling distribution; two-sided mean tests recenter the paired difference under the null. These procedures assess the realized strategy-return comparison under the maintained time-series approximation. They do not re-estimate the factor model or implement the companion paper’s loading-report inference. Table IV applies the procedure to QZ-IPCA versus IPCA; Tables VI and VII apply it to the intercept inputs. Table IA.VI varies block length, and Table IA.II reports the shorter common-sample metric comparison. The resampling statistics and seeds are specified in Internet Appendix IA.5.3.

Intervals are pointwise for the stated comparison, not joint confidence statements over all specifications. Where reported, Holm-adjusted p -values control multiplicity across the four capitalization universes within one panel; the underlying return records remain jointly dependent.

The replication package supports two complementary checks: rebuilding reported statistics from frozen monthly series with recorded provenance, and examining the risk-matrix intervention using saved mechanism diagnostics. These provide separate routes for checking reported performance and the stored-matrix mechanism. Full upstream validation additionally requires checks of information dates, stock selection, and optimizer conditions. Section VI.D specifies the remaining checks needed for a fully validated investment result.

III. The Empirical Ruler, Forecasts, and Portfolios

A. *Concentrated exposures and a changing empirical ruler*

We begin with the geometry actually used in the EWMA comparison, rather than a separate calibration. Figure 1 uses all 618 saved formation-date Gram matrices. All have full rank, but their eigenvalue mass is concentrated. The spectral participation dimension is $(\text{tr } \widehat{Q}_t)^2 / \text{tr}(\widehat{Q}_t^2)$: it equals 132 if the eigenvalues are equal and falls as their mass becomes more concentrated. Its median is 19.8 and its range is 17.4–22.6. The leading eigenvalue accounts for a median 14.2% of the trace. Hence a full-rank library can contain much less dispersed exposure variation than its nominal size suggests. Neither statistic estimates the number of independent or priced risks.

The normalized spectrum changes gradually. Median month-to-month spectral drift is 0.063 percentage points and its maximum is 0.115 percentage points; Table IA.III gives the full summary and (IA21) defines the diagnostic. This is consistent with a long, overlapping estimation window and a 60-month EWMA half-life. Participation summarizes concentration, not the overall exposure scale or which characteristic combinations carry it. A nearly constant participation dimension can therefore coexist with a changing metric. The two panels separate the concentration level from adjacent changes in the spectrum. NBER bands provide historical context for these spectral summaries, not evidence of a contemporaneous recession response. Preprocessing changes in every adjacent window, so the path combines changes in the sample, transformations, and characteristic relationships rather than isolating market-state shocks.

The investment question is therefore more specific than whether a richer set of characteristics improves fit. Does applying a different ruler to these exposures change the portfolio, and through which input? The next comparison holds the forecast nearly fixed; the subsequent risk-matrix intervention isolates the role of the exposure penalty.

B. *The matched exponentially weighted comparison*

We first check whether separately estimated models approximately reproduce the forecast agreement that a matched representation would preserve exactly. In the EWMA metric experiment, the median relative Euclidean difference between the QZ-IPCA and IPCA forecast vectors is 4.7×10^{-6} . The 95th percentile is 1.8×10^{-5} and the maximum is 1.4×10^{-4} . These diagnostics use the forecast

Gray: NBER recession; white: expansion

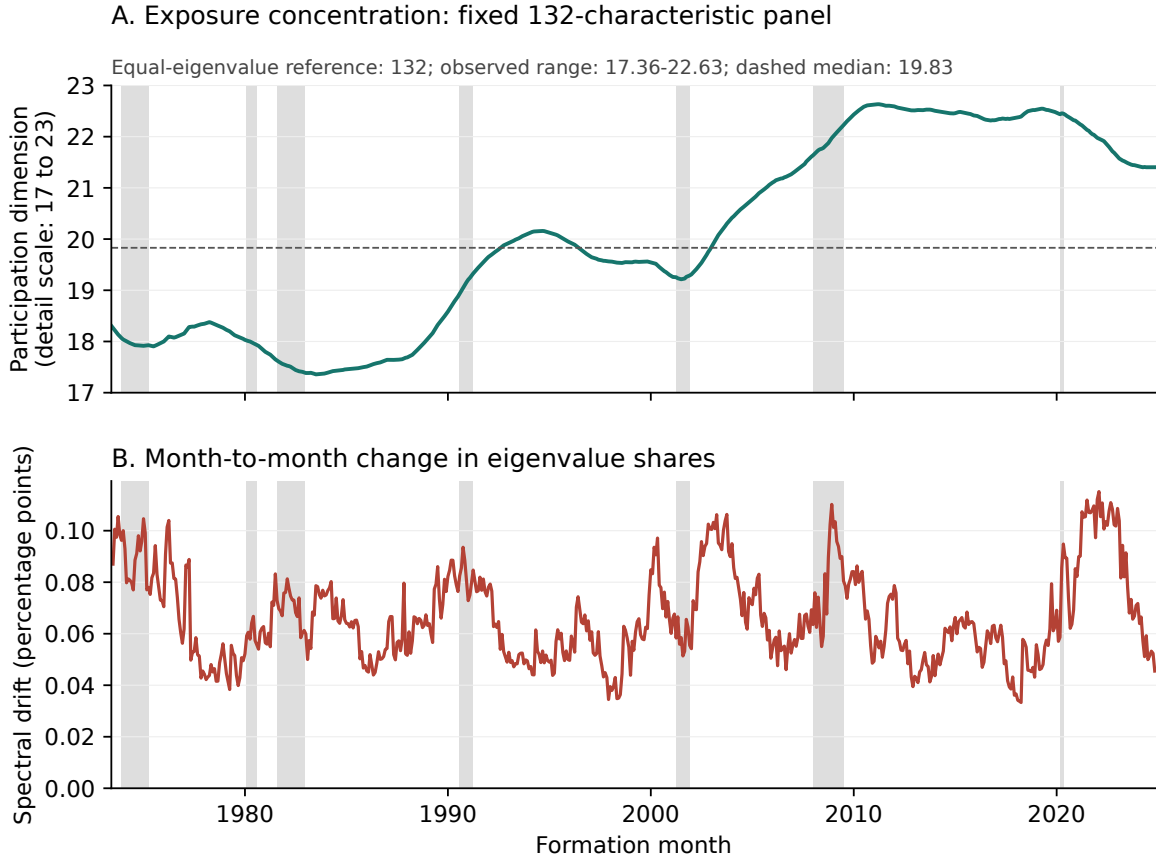


Figure 1. The rolling empirical ruler on the production sample. Panel A reports the spectral participation dimension of the 132-feature EWMA Gram matrix in each of 618 formation months, June 1973–November 2024. Its detail axis spans 17–23; the dashed line is the sample median and the equal-eigenvalue benchmark is 132, outside this axis. Panel B reports one half the absolute change in its normalized ordered eigenvalue shares, summed across all 132 eigenvalues, in percentage points. This scale-normalized spectral drift describes concentration changes; it does not measure rotations of eigenvectors. The matrices incorporate each window’s preprocessing and stock sample. The figure is reconstructed from saved matrices, not from a new estimation of stock returns. Gray bands mark NBER recession months on the formation-date axis; unshaded periods are expansions.

vectors underlying the risk-matrix comparison, with the relative norm defined in (IA19); they are measured numerical agreements, not imposed equality.

The associated portfolio weights are not nearly as close. Their median cross-model cosine is 0.83, with interquartile range [0.753, 0.878]. Positive leverage normalization leaves this cosine unchanged, so the gap concerns the direction of the portfolio rather than just its scale. The mechanism analysis therefore starts with a precise discrepancy: approximately the same expected-return input enters two portfolio solves, but the resulting relative stock positions differ materially.

This comparison makes portfolio allocation a distinct object of model evaluation: nearly unchanged forecasts coexist with materially different relative stock positions. Forecast statistics alone therefore leave an economically relevant implementation choice unresolved. Section IV identifies the risk-matrix term responsible for this weight separation within the stored estimates.

C. Realized performance under the recorded metrics

Table III reports performance levels and Table IV reports paired differences and uncertainty. EWMA uses the same metric as the mechanism diagnostics. On the 618-month sample, its annualized mean is 4.6%, volatility is 5.1%, and gross Sharpe ratio is 0.91, compared with 0.64 for IPCA. The newly computed Sharpe-difference interval is $[-0.02, 0.58]$; its mean-return test has $p = 0.25$. Thus the same-line point estimates favor QZ-IPCA, but the intervals do not establish a gain at the 5% level.

The 120-month equal-weight metric has a Sharpe ratio of 0.85 and a Sharpe-difference interval of $[-0.06, 0.51]$ relative to IPCA. Its mean-return test has $p = 0.32$. The recorded rolling-mean specification provides a third comparison. On the common 618-month sample, QZ-IPCA earns an annualized gross mean of 4.8% at 4.8% volatility, against 3.9% at 6.1% for IPCA. Their Sharpe ratios are 1.00 and 0.64, respectively. Their compounded maximum drawdowns are -16.5% and -34.6% . All figures refer to the unit-gross policy.

For the rolling-mean specification, the paired bootstrap Sharpe-ratio difference is 0.37, with 95% interval $[0.02, 0.68]$. The difference uses unrounded series and therefore need not equal the difference of the printed levels. The mean-return difference is 7.9 basis points per month; its separately recorded mean test gives a 95% interval of $[-3.6, 19.5]$ basis points and $p = 0.17$. Volatility and drawdown difference intervals also include zero. The composite Sharpe comparison is supported by this resampling exercise, while a separate mean gain or volatility reduction is not established at the same confidence level. The rolling-metric Sharpe interval widens to $[-0.01, 0.69]$

Table III. Gross portfolio performance by metric construction. The Core sample has 618 months, June 1973–November 2024. All strategies use unit-gross weights. Means and volatilities are annualized; drawdowns are compounded. EWMA uses the 120-month window with a 60-month half-life; the equal-weight average uses all 120 months; the rolling mean uses its final 60 months. The forecast and ridge-removal diagnostics concern EWMA. Entries are recalculated from the recovered monthly series and rounded once.

Statistic	IPCA	QZ-IPCA		
		EWMA	Equal 120	Rolling 60
Gross Sharpe ratio	0.64	0.91	0.85	1.00
Annualized mean (%)	3.9	4.6	4.5	4.8
Annualized volatility (%)	6.1	5.1	5.3	4.8
Maximum drawdown (%)	-34.6	-18.0	-19.8	-16.5

Table IV. Paired performance inference, QZ-IPCA minus IPCA. The Core sample is June 1973–November 2024 (618 months); returns are unit-gross and gross of costs. The revision uses paired 12-month blocks, 10,000 draws, seed 20260714, and pointwise percentile intervals for all three metric constructions. The separately recentered mean test uses seed 20260713. Different seeds produce slightly different mean-interval endpoints. These tests compare each metric with IPCA, not the metrics with each other; they are not adjusted for metric selection. Drawdown is a path diagnostic, not an inferential claim.

Difference	Estimate	95% interval
EWMA, 120 months		
Annualized gross Sharpe	0.27	[−0.02, 0.58]
Annualized mean (percentage points)	0.76	[−0.45, 2.12]
Annualized volatility (percentage points)	−1.00	[−2.32, 0.18]
Recentered mean-test $p = 0.25$		
Equal 120		
Annualized gross Sharpe	0.21	[−0.06, 0.51]
Annualized mean (percentage points)	0.64	[−0.54, 1.96]
Annualized volatility (percentage points)	−0.76	[−2.06, 0.37]
Recentered mean-test $p = 0.32$		
Rolling 60		
Annualized gross Sharpe	0.37	[0.02, 0.68]
Annualized mean (percentage points)	0.95	[−0.38, 2.36]
Annualized volatility (percentage points)	−1.29	[−2.62, 0.04]
Recentered mean-test $p = 0.17$		

with 24-month blocks and $[-0.01, 0.76]$ with 60-month blocks. Thus its exclusion of zero depends on the 12-month approximation. The EWMA and equal-120 intervals include zero at all three lengths. The complete stress check appears in Table IA.VI.

Rolling-60 QZ-IPCA has a higher gross Sharpe ratio in each of three fixed 206-month blocks: 0.93 versus 0.69 in June 1973–July 1990, 1.88 versus 1.75 in August 1990–September 2007, and 0.41 versus -0.14 in October 2007–November 2024. This is descriptive evidence of persistence across periods, not three additional independent tests. The recent block also shows a substantial decline in performance for both rules.

Across the three recorded constructions, available-history QZ-IPCA Sharpe ratios range from 0.85 to 1.00, allowing investors to compare the portfolio consequences of declared exposure conventions. Reporting the full set keeps this comparison separate from retrospective selection by significance. The original significant Sharpe comparison belongs to the rolling-mean line and remains subject to the block-length sensitivity above; the EWMA comparison does not share that significance result. Because all three metrics update, these comparisons assess alternative updating rules rather than isolate the value of updating itself. That question requires an otherwise matched frozen-metric policy and consistently transported portfolio inputs.

IV. How a Normalization Becomes an Exposure Penalty

A. An exact decomposition of portfolio differences

Let A_I, A_Q denote the positive-definite asset risk matrices and μ_I, μ_Q the two expected-return vectors at a fixed formation date; subscripts I and Q refer to IPCA and QZ-IPCA, respectively, and time subscripts are suppressed. Adding and subtracting $A_Q^{-1}\mu_I$ and using the inverse-matrix identity gives

$$w_Q - w_I = A_Q^{-1}(\mu_Q - \mu_I) + A_Q^{-1}(A_I - A_Q)w_I. \quad (11)$$

The first term changes the mean input at the QZ risk matrix; the second changes the risk matrix at the IPCA mean input. This is an exact decomposition, not a regression attribution.

In the exponentially weighted experiment, the first term's norm has median 0.07% of the norm of the total weight difference. The two term norms are not additive shares, because the terms need not be orthogonal. Nevertheless, the small mean term directs attention to the covariance input.

The idiosyncratic diagonals are very close: their median relative gap is 4.2×10^{-7} . The largest principal angle between the latent loading spans has median 4.3×10^{-4} degrees and maximum 0.011 degrees. These diagnostics point to the treatment of factor risk, rather than a large change in fitted latent span or forecast mean.

B. The factor-coordinate ridge

Consider two matched factor representations related by a nonsingular matrix $J \in \mathbb{R}^{(K+5) \times (K+5)}$, with $B_Q = B_I J^{-1}$, $m_Q = J m_I$, and $C_Q = J C_I J'$. Here C_I and C_Q are factor covariances before the ridge is added. For the no-intercept case of Proposition 1, the joint transformation is

$$J = \begin{pmatrix} C^{1/2} & C^{1/2} D \\ 0 & I_5 \end{pmatrix}.$$

The transported mean and unregularized stock covariance satisfy

$$B_Q m_Q = B_I m_I, \quad B_Q C_Q B_Q' = B_I C_I B_I'. \quad (12)$$

These identities follow by cancellation of J and J^{-1} . They require matched covariance and forecast constructions as well as matched loadings.

Adding the same factor-coordinate ridge to both representations instead gives

$$B_Q (C_Q + \lambda_F I) B_Q' = B_I C_I B_I' + \lambda_F B_Q B_Q'. \quad (13)$$

The corresponding term in the other representation is $\lambda_F B_I B_I'$. A general nonsingular change of factor basis does not preserve these two matrices. Orthogonal changes do; a change in metric normalization need not be orthogonal.

This observation has a direct portfolio interpretation. With $A_0 = B \widehat{C}_F B' + \widehat{D} + \lambda_\Sigma I$, where the subscript zero denotes removal of the factor ridge only, the implemented solve maximizes

$$w' \widehat{\mu} - \frac{1}{2} w' A_0 w - \frac{\lambda_F}{2} \|B' w\|^2. \quad (14)$$

The ridge penalizes the portfolio’s factor exposures in their reported coordinates. For example, rescaling a factor upward rescales its stock loadings downward. An unchanged numerical ridge then imposes a smaller penalty on that coordinate for any given stock portfolio, even when the transported forecast and unregularized stock covariance are unchanged. The optimizer can consequently choose different relative stock positions. Holding the ridge fixed across normalizations therefore specifies an exposure preference as well as a numerical stabilization rule.

The implication is constructive. A factor-risk adjustment R preserves the matched stock covariance if it is transported as $R_Q = JR_IJ'$. Alternatively, a directly specified stock-space adjustment can be held common across representations. These designs retain the benefits of regularization while making the induced exposure preference explicit. They do not require choosing the empirical metric by in-sample return fit, which is invariant under a matched representation change.

C. The removal experiment and covariance reconciliation

The stored risk matrices allow the channel to be removed without changing the forecast, the stock residual diagonal, or the asset-space ridge. For every month, we subtract $\lambda_F I$ from the stored factor covariance and solve the two portfolios again. Table V reports the resulting diagnostics.

The median value of one minus the weight cosine falls from approximately 0.17 to 6×10^{-12} ; its maximum is 4.6×10^{-9} . Cosine agreement alone would only establish proportionality. The relative weight difference also falls to a median of 3.9×10^{-6} and a maximum of 9.6×10^{-5} , supporting near equality of the raw solves, not merely their directions. Figure 2 displays the month-by-month diagnostic.

A second calculation reconstructs the factor covariances from the saved latent factors and observed FF5 history. Production pairs characteristics dated s with returns and observed factors realized at $s + 1$. An earlier diagnostic instead selected FF5 observations at s , creating an apparent covariance discrepancy and omitting the first available window. Correcting that alignment reproduces the stored covariance less 0.01% in all 618 windows of each model. Across the 1,236 matrices, the maximum relative Frobenius discrepancy is below 2×10^{-15} , at machine precision.

Within the saved EWMA estimates, removing the factor-coordinate ridge nearly eliminates the cross-model weight gap, linking the observed allocation differences to the implemented exposure

Table V. EWMA mechanism diagnostics. The Core sample has 618 months, June 1973–November 2024. Correctly dated factor histories reconstruct all 618 covariances in each model. Norm ratios are diagnostic magnitudes, not additive shares. Cosines are unchanged by positive leverage rescaling. The removal row deletes only the factor-coordinate ridge from the saved matrices.

Diagnostic	Median	Maximum or minimum
Mean-term norm / total weight-gap norm (%)	0.07	—
Relative idiosyncratic-variance gap	4.2×10^{-7}	—
Largest latent-span angle (degrees)	4.3×10^{-4}	Max. 0.011
Implemented weight cosine	0.830	—
Relative raw-weight gap before removal	0.607	Max. 0.895
1–cosine after ridge removal	6.0×10^{-12}	Max. 4.6×10^{-9}
Relative raw-weight gap after removal	3.9×10^{-6}	Max. 9.6×10^{-5}
Rebuilt covariance relative gap, both models	—	$< 2 \times 10^{-15}$

The covariance row compares the covariance of saved latent factors and observed factors paired with the same return dates, against the stored matrix less $\lambda_F I = 0.01I$. It covers 1,236 model-window matrices. The first three diagnostics correspond to the mean and covariance channels in (11); none is a percentage contribution to realized returns.

penalty. Correctly dated factor histories independently reproduce the ridge-removal input, checking covariance construction from saved estimates rather than re-estimating the stock panel. Attributing the realized Sharpe difference would additionally require return paths for ridge-free or transported-ridge portfolios, which have not been evaluated here. The weight result gives covariance audits a concrete starting point: examine factor units and regularization before interpreting allocation differences as new forecasting information or a newly identified source of risk.

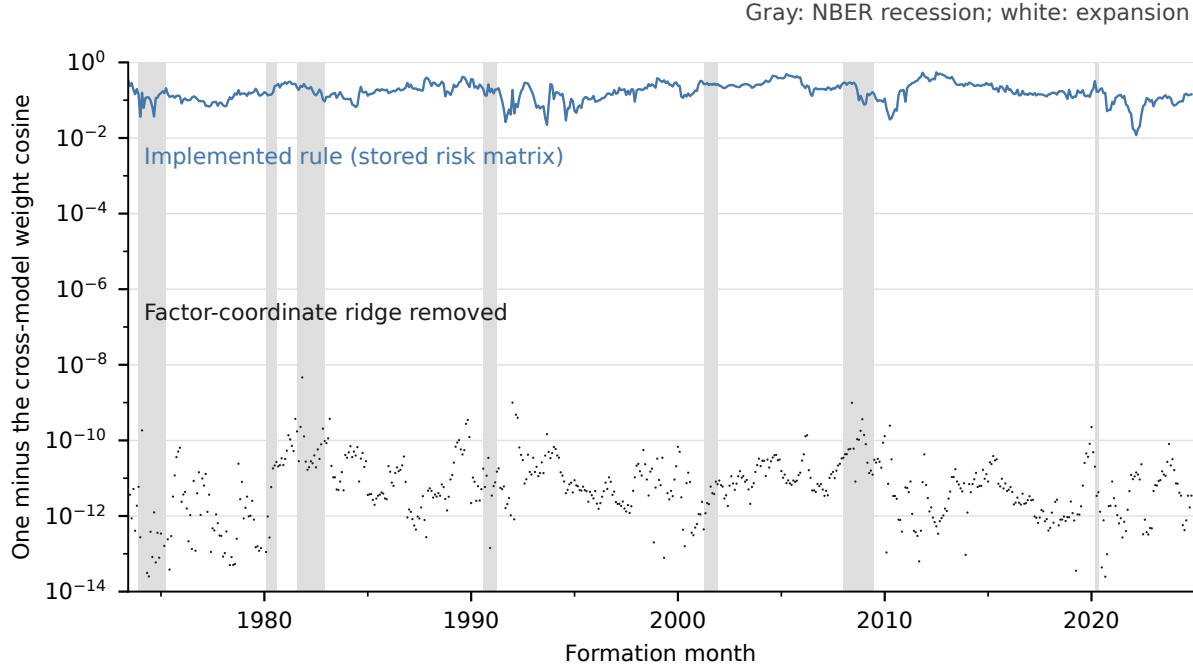


Figure 2. The factor-coordinate ridge removal diagnostic. The exponentially weighted QZ-IPCA and IPCA comparison uses the 618 available-history months. The figure compares one minus the cross-model portfolio-weight cosine before and after deleting the factor-coordinate ridge from the stored risk matrices. Values near numerical zero reflect agreement of directions; the relative-weight diagnostics in the text additionally compare scale. Nine ridge-removed observations below the 10^{-14} plotting floor are omitted, including five with a recorded cosine of one. Gray bands mark NBER recessions at formation dates; unshaded periods are expansions. This is a stored-matrix intervention, not a raw-panel re-estimation.

V. What the Intercept Adds to the Portfolio

A. An input-level comparison within one fitted model

The matched no-intercept comparison holds expected returns approximately fixed. The intercept-inclusive extension asks a different question: what happens when a retained block is included in the portfolio's mean input? Using the same NA-IPCA fit, define

$$\mu_t^{\text{IAB}} = B_t \widehat{m}_{F,t} + Z_t \widehat{\Gamma}_{\alpha,t}. \quad (15)$$

The second term restores the characteristic-dependent intercept under the declared metric convention. It makes explicit the part of the fitted mean that a factor-only portfolio omits. Using the intercept asks whether this retained return signal helps allocate capital after accounting for the portfolio's risk, rather than whether it can be relabelled as a factor.

The implemented rule also adds a positive-semidefinite loading-instability term to its risk matrix:

$$\Sigma_t^{\text{IAB}} = B_t \Sigma_{F,t} B_t' + \widehat{D}_t + \kappa_U Z_t V_{\Gamma_{\alpha,t}} Z_t', \quad \kappa_U = 1, \quad (16)$$

$$w_t^{\text{IAB}} = (\Sigma_t^{\text{IAB}} + \lambda_{\Sigma} I)^{-1} \mu_t^{\text{IAB}}. \quad (17)$$

The scalar κ_U is the loading-instability multiplier. On the strict-24 sample, $V_{\Gamma_{\alpha,t}}$ is the sample covariance of the 24 most recent prior intercept-loading estimates. It has rank at most 23 in 132 characteristic dimensions. Overlapping estimation windows induce dependence, and changes in loadings may combine estimation error with underlying parameter movement. Window-specific characteristic standardization can also change the units of the saved coefficients. We therefore interpret this input as an instability proxy. It becomes a conditional predictive-risk adjustment only under additional assumptions linking it to forecast estimation error; $\kappa_U = 1$ is a specification choice, not an established optimum.

Three portfolios distinguish the channels. The intercept-off ablation removes both $Z_t \widehat{\Gamma}_{\alpha,t}$ and the instability term from the same estimated model. The mean-only version restores the intercept but sets $\kappa_U = 0$. The complete IAB rule retains both. This comparison identifies the contribution of portfolio inputs, rather than the performance gain from estimating a different model.

The mean channel follows directly from the portfolio normal equations. Let w_t^{off} and w_t^{mean} denote the unnormalized intercept-off and mean-only solutions. Holding the risk matrix A_t in (9) fixed within this fit, subtraction gives

$$w_t^{\text{mean}} - w_t^{\text{off}} = A_t^{-1} Z_t \widehat{\Gamma}_{\alpha,t}. \quad (18)$$

The intercept changes the expected-return input; the inverse risk matrix converts that signal into a joint reallocation across stocks. This increment need not be factor-neutral: neutrality to the fitted factor exposures would require $B_t' A_t^{-1} Z_t \widehat{\Gamma}_{\alpha,t} = 0$. Characteristic-space orthogonality in (2) does not impose that stock-portfolio restriction.

The instability channel instead changes the quadratic cost of an exposure. In the mean–variance objective, its added penalty is

$$\frac{\kappa_U}{2} w_t' Z_t V_{\Gamma_{\alpha,t}} Z_t' w_t = \frac{\kappa_U}{2} (Z_t' w_t)' V_{\Gamma_{\alpha,t}} (Z_t' w_t). \quad (19)$$

The mean input rewards exposure to the fitted intercept. The instability penalty is proportional to the sample variance of that exposure’s intercept contribution across the recorded prior loading estimates, holding current stock weights and characteristics fixed. A position can therefore offer a large fitted mean contribution while receiving a large instability penalty. The two inputs distinguish the return being forecast from the recorded loading variation borne by the position. These identities describe raw portfolio directions. Each rule is subsequently normalized by its own gross exposure, so their realized returns must be compared directly rather than added as separate return contributions.

The intercept-off rule provides a metric-specific control within the NA-IPCA fit. Under the matched transformation in Proposition 1, the factor-only fitted mean gains $Z_t \Gamma_{\beta} a$, exactly offsetting the reduction in the intercept contribution. Only their sum is preserved. Omitting the intercept after fitting thus tests the use of one fitted component; re-estimating under $\Gamma_{\alpha} = 0$ tests a different model. The same-fit comparison isolates the portfolio-input channel, while the separately fitted QZ-IPCA benchmark evaluates the full specification. Keeping both comparisons makes their economic roles explicit.

B. Performance and the channel carrying the difference

The paired gross-return and Sharpe comparisons measure the investment value of using the fitted intercept. They complement the model interpretation in Section I.C: benchmark-pricing alpha would answer the separate question of performance relative to specified risk exposures. On the strict-24 Core sample, the complete rule has gross annualized mean 7.60%, volatility 3.00%, and Sharpe ratio 2.53, against 1.75%, 4.46%, and 0.39 for the intercept-off ablation. Its paired mean-return increment is 0.488% per month, with a 95% block-bootstrap interval of [0.348, 0.634]%. Table VI reports the capitalization-specific comparisons. The model and intercept history are estimated on Core stocks. The same loadings and Core standardization are then applied to Mega, Large, and Small stocks, and portfolios are re-solved and normalized within each group. These

are dependent applications of a common fit, which permit comparison across capitalization groups while holding the estimated loading model fixed. All four full-sample mean tests remain significant at 1% after the reported Holm adjustment.

Table VI. Paired mean-return increment of complete IAB over its intercept-off ablation. Returns use unit-gross weights and are gross of costs. Means and intervals are in percent per month. The 12-month moving-block bootstrap uses 10,000 draws, seed 20260714, and a recentered two-sided mean test. Holm adjustment covers the four universes within each panel, not the broader set of research specifications. All rows use the EWMA NA-IPCA fit estimated on Core, then applied and re-solved within each indicated stock group with its own unit-gross normalization.

Universe	Mean	95% interval	p	Holm p
Full strict-24 sample: June 1975–November 2024 (594 months)				
Core	0.488	[0.348, 0.634]	< 0.001	< 0.001
Mega	0.239	[0.105, 0.409]	0.003	0.003
Large	0.347	[0.194, 0.508]	< 0.001	< 0.001
Small	0.666	[0.487, 0.840]	< 0.001	< 0.001
Recent period: October 2007–November 2024 (206 months)				
Core	0.202	[−0.071, 0.409]	0.11	0.32
Mega	0.206	[−0.035, 0.507]	0.14	0.32
Large	0.174	[−0.120, 0.389]	0.19	0.32
Small	0.251	[−0.061, 0.464]	0.08	0.32

The recent period gives a more restrained picture. From October 2007 to November 2024, the final 206-month block of the available-history sample, the Core mean difference is 0.202% per month, with interval $[-0.071, 0.409]\%$ and $p = 0.11$. None of the recent-period intervals excludes zero. The recent sample therefore gives less precise evidence about the mean increment, an important consideration for prospective use. Establishing a change across periods requires a direct between-period test.

Table VII separates the two inputs on the same 594 months. Restoring the intercept in the mean alone raises the Sharpe ratio from 0.3917 to 2.5126. The complete rule raises it further to 2.5347. The same-fit, same-sample design locates the large difference in the restored mean input, before the instability adjustment is introduced.

The smaller second increment is measurable at its own scale. The complete rule raises the Sharpe ratio by 0.0222 relative to mean-only, with pointwise interval $[0.0100, 0.0355]$. The monthly mean

Table VII. Which portfolio input carries the intercept result? All entries use the same NA-IPCA estimates, EWMA metric, 594 formation months (June 1975–November 2024), and unit-gross policy. Off removes the intercept from both mean and risk inputs; mean-only restores it in the mean; complete additionally uses the trailing-loading instability proxy with $\kappa_U = 1$. Panel B reports pointwise 95% paired 12-month-block intervals from 10,000 draws using seed 20260714; recentered mean tests use seed 20260713. No adjustment for specification selection is made. This is an input ablation, not a comparison of separately fitted models.

Panel A: Gross performance			
Statistic	Off	Mean-only	Complete
Annualized mean (%)	1.75	7.59	7.60
Annualized volatility (%)	4.46	3.02	3.00
Sharpe ratio	0.3917	2.5126	2.5347
Maximum drawdown (%)	-17.73	-3.93	-3.87
Panel B: Incremental input effects			
Comparison	Monthly mean (bp)	95% interval	<i>p</i>
Mean-only minus off	48.69	[34.66, 63.36]	< 0.001
Complete minus mean-only	0.08	[-0.15, 0.26]	0.437
Comparison	Sharpe difference	95% interval	
Mean-only minus off	2.1209	[1.6445, 2.6791]	
Complete minus mean-only	0.0222	[0.0100, 0.0355]	

difference is only 0.08 basis points, with interval $[-0.15, 0.26]$ and $p = 0.44$. Annualized volatility falls by 0.0226 percentage points, with pointwise interval $[-0.0391, -0.0084]$ percentage points for complete minus mean-only. Thus the proxy contributes a smaller risk-profile adjustment on top of the large mean-input effect. This distinction identifies which portfolio input carries each measured change. Internet Appendix IA.7.2 reports the wider multiplier sweep and the distinct full-rank sensitivity exercise, neither of which determines the specification reported here.

The positive Sharpe intervals for both input increments remain positive under 24- and 60-month blocks, whereas the small mean-return increment from the risk adjustment continues to include zero. The mean-versus-risk interpretation is therefore preserved across these declared block lengths. Section VI.D addresses validation of the upstream forecast and instability inputs.

The evidence directs attention first to the retained mean input and then to its smaller risk adjustment. This is an input attribution within one fitted model. We next compare complete, separately estimated strategies.

VI. Common-Sample Performance and Economic Interpretation

A. Strategy comparisons on a common sample

The preceding input ablation asks how investors use one fitted model. Comparing separately estimated strategies asks a complementary question: how do the return and risk estimates supplied by alternative models perform under the same allocation rule? Table VIII places these strategies on the 594 common formation months from June 1975 to November 2024. This strict-24 sample gives the intercept-instability adjustment a full history of 24 prior loading estimates at every formation date. All strategies use unit-gross weights, so differences reflect the positions selected under a common gross-exposure budget rather than different evaluation periods or gross-exposure targets.

The complete IAB rule has an annualized gross Sharpe ratio of 2.53, compared with 1.49 for RP-PCA, 0.75 for FF5, and 0.69 for IPCA. The three QZ-IPCA specifications yield 1.07 under EWMA, 1.03 under the 120-month equal-weight metric, and 1.13 under the rolling-60 metric. These levels describe complete investment rules. The same-fit comparison in Table VII provides the more specific attribution: most of the IAB improvement over its intercept-off control comes from restoring the intercept to the expected-return input.

Table VIII. Separately estimated strategies and a within-fit control. The strict-24 sample is June 1975–November 2024 (594 months). All returns use unit-gross weights and are gross of trading costs. Volatility is annualized; drawdown is compounded. All three recorded QZ-IPCA constructions are included. The intercept-off ablation uses the same NA-IPCA estimates and is not a separately estimated comparator. The degenerate archived PCA configuration is excluded. IAB and its intercept-off control use EWMA.

Specification	Sharpe	Volatility (%)	Drawdown (%)
FF5	0.75	5.2	−20.7
RP-PCA	1.49	3.9	−13.4
IPCA	0.69	5.9	−34.6
QZ-IPCA, EWMA	1.07	4.5	−14.2
QZ-IPCA, equal 120	1.03	4.6	−17.3
QZ-IPCA, rolling 60	1.13	4.3	−11.1
NA-IPCA, complete IAB	2.53	3.0	−3.9
NA-IPCA, intercept-off ablation	0.39	4.5	−17.7

A common gross-exposure budget does not impose a common risk profile. Annualized volatility ranges from 3.0% to 5.9%, and drawdowns reflect the different stock positions generated by

each model and allocation rule. The comparison therefore evaluates forecast and risk inputs jointly. Table IA.II reports paired differences between each QZ-IPCA specification and IPCA. All three pointwise 95% Sharpe-difference intervals exclude zero with 12-month blocks, while the corresponding mean-return intervals include zero. The evidence supports differences in the ratio of mean return to volatility under this resampling design, rather than a separately established increase in mean return. The EWMA and equal-120 Sharpe intervals include zero on the longer 618-month sample. Keeping both samples visible distinguishes the portfolio mechanism from the precision of its realized-performance comparisons.

Figure 3 shows when the performance differences accumulate. Each path sums monthly unit-gross returns after dividing by the strategy's full-sample standard deviation. The endpoint is proportional to its full-sample Sharpe ratio, whereas the intermediate path shows the timing of standardized gains and losses. This retrospective scaling is a display convention, not an additional volatility-targeting policy.

B. Forecast accuracy and investment value

Forecast statistics and portfolio returns answer different questions. Table IA.IV evaluates stock-level predictions on the 594-month common sample. IPCA and rolling-60 QZ-IPCA agree to four decimals in both predictive R^2 and cross-sectional association. RP-PCA has the highest rank correlation among the reported forecasts, but its predictive R^2 relative to zero is approximately -0.26 . It nevertheless has the highest gross Sharpe ratio among the separately estimated benchmarks other than complete IAB.

The distinction follows from the objectives being evaluated. Squared forecast errors depend on both the pattern and magnitude of predicted returns. Portfolio weights depend on those forecasts relative to the risk matrix and the chosen exposure budget. For example, holding the risk matrix fixed, multiplying a nonzero forecast vector by a positive constant leaves its unit-gross mean-variance portfolio unchanged, although its squared prediction error generally changes. Correlation describes cross-sectional association without assessing forecast scale. The RP-PCA results illustrate why these measures are complementary: a forecast can help order stocks for investment while performing poorly as a prediction of return levels.

Gray: NBER recession; white: expansion

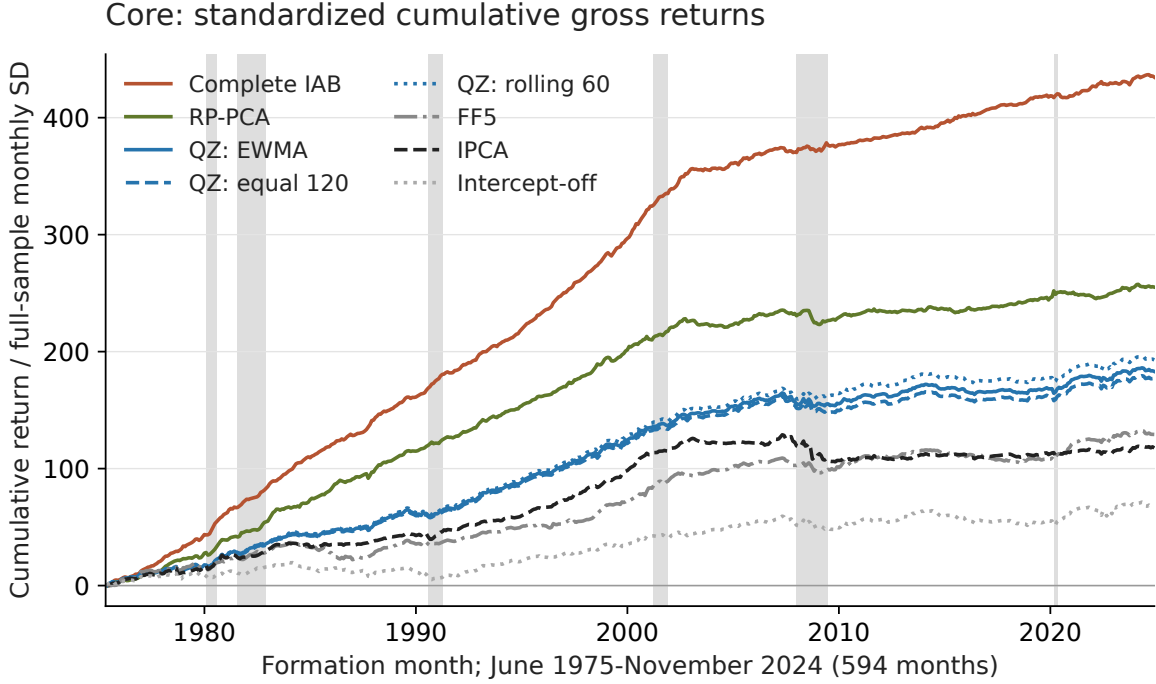


Figure 3. Standardized cumulative gross returns on the common Core sample. For strategy m , the path is $S_m(\tau) = \sum_{t=1}^{\tau} r_{m,t+1}^u / \hat{\sigma}_m$, where $\hat{\sigma}_m$ is its monthly sample standard deviation over all $T = 594$ formation months, June 1975–November 2024. Returns use unit-gross weights and are before trading costs. The identity $S_m(T) = T\widehat{SR}_m / \sqrt{12}$ links the endpoint to Table VIII. All three QZ-IPCA metric constructions are shown. Intercept-off uses the same NA-IPCA fit as complete IAB. Paths are cumulative sums, not compounded wealth or rolling Sharpe ratios. The fixed display scaling leaves the underlying time-varying unit-gross policy unchanged. Gray bands mark NBER recession months on the formation-date axis.

For NA-IPCA, the table evaluates the factor-only forecast, whereas the complete IAB mean also contains the estimated intercept. Its portfolio result therefore concerns the value of allocating on that additional mean component under the specified risk rule. The table does not evaluate the calibration of the complete IAB forecast. This distinction keeps the portfolio-input finding separate from a claim about stock-level squared-error improvement.

C. Sensitivity of the allocation channels

The common-initialization comparison examines whether the forecast–weight contrast persists when starting values are held fixed. Table IA.I uses a common start, no block-loading penalty, and the production moment ridge. The EWMA comparison again produces nearly matching latent spans while portfolio weights remain different. This supports the representation and regularization channel identified in Section IV, rather than attributing the contrast solely to different initial loading spans.⁴

A separate re-expression exercise isolates the loading-instability input. Across nine recorded conventions on the 618-month sample, the $\kappa_U = 1$ gross Sharpe ratios range from 2.4895 to 2.4938. The mean forecast and systematic covariance are held fixed, and only the instability adjustment is re-expressed. The narrow range shows that this particular risk-input change has little effect on the complete portfolio’s gross performance. Equality at $\kappa_U = 0$ follows from the design. The exercise thus measures sensitivity of one allocation input, rather than performance of fully re-estimated strategies under alternative metrics. Internet Appendix IA.7.2 also reports multiplier and alternative-matrix comparisons.

Time variation is more pronounced. The complete IAB Sharpe ratios are 3.97, 3.03, and 1.13 in the three consecutive 198-month blocks of the strict-24 sample. The lower final-block performance makes the timing of the full-sample result economically relevant. The recent-period mean comparison in Section V.B is also less precise than the full-sample comparison; these estimates do not by themselves establish a structural change in expected returns. Internet Appendix IA.7.5 reports the subperiod design and rolling comparisons. A separate diagnostic removes each strategy’s five largest absolute-return months. The IAB Sharpe remains near 2.53, while the IPCA and FF5 ordering reverses. This assesses sensitivity to extreme observations within each strategy; because the retained dates differ, it is not a paired comparison on a new common sample.

⁴The June 1999 window reaches the iteration cap in both specifications. Internet Appendix IA.1.3 reports the stopping criterion and constraint diagnostics; common initialization does not establish stationarity in every window.

D. Economic interpretation and implementation scope

The evidence distinguishes the economic roles of exposure measurement, return prediction, and portfolio policy. The matched-model comparison locates an allocation difference in the factor-coordinate penalty; the same-fit intercept comparison locates a performance difference primarily in the mean input. Neither interpretation requires treating statistical concentration as a priced risk. A pricing interpretation of the retained intercept instead depends on the factor-pricing benchmark developed in Section I.C.

The empirical performance estimates concern gross returns under the stated unit-gross policy. Net investment value also depends on trades from return-drifted holdings to the next normalized portfolio, as well as transaction, borrowing, and financing costs. The cost proxy in Internet Appendix IA.4.4 does not reconstruct those trades and therefore does not estimate implementable net performance or capacity. The return comparisons are also conditional on the processed panel's eligibility, information-date, and return conventions. The saved-factor reconciliation checks covariance construction, while full point-in-time availability of the underlying inputs remains unverified; Internet Appendix IA.1.4 records the relevant data conventions.

The companion theory explains inference for loading classes under its local identification and estimator-accuracy conditions. The portfolio instability matrix has a different role: it penalizes directions in which past intercept-loading estimates have varied. That variation can combine estimation error, parameter movement, and changes in characteristic units. It is consequently an instability input, rather than a calibrated sampling covariance for the constrained estimator. The gross-return comparisons evaluate its use as part of the declared EWMA policy; they do not establish metric-independent performance of the intercept-inclusive strategy.

Finally, all three characteristic metrics update over time. Their comparison measures the consequences of alternative exposure conventions, not the incremental value of updating itself. An otherwise matched frozen-metric policy, with characteristic units held fixed and risk adjustments consistently transported, would isolate that adaptation question. The present comparisons establish the prior distinction needed for such an analysis: a change in measured geometry can alter portfolio preferences even when fitted return information is preserved.

VII. Conclusion

Characteristic-based factor models provide return forecasts, but their investment implications also depend on how portfolio exposures are measured and penalized. We show that a normalization can enter this decision through an otherwise fixed factor-coordinate ridge. Matched representations preserve complete fitted returns and, with consistently transformed factor moments, stock-level mean and unregularized risk inputs. A penalty held fixed in factor coordinates can nevertheless change the stock positions selected by the investor.

The rolling U.S. equity application traces this channel in a model with 132 characteristics. The Gram metric measures overlap in the stock-level exposures generated by those characteristics. In the EWMA comparison, removing the factor-coordinate ridge nearly eliminates the weight gap between models with almost identical forecasts. This identifies the allocation channel within that comparison. Common-sample gross returns then evaluate the resulting investment rules, with statistical precision depending on the metric, evaluation period, and resampling design.

The intercept extension addresses a separate portfolio choice: whether to use a fitted mean component retained outside the factor blocks. Within the same NA-IPCA fit, most of the improvement over the intercept-off portfolio comes from including that component in expected returns; the loading-instability adjustment contributes a smaller increment. The finding locates the investment contribution in the use of the fitted signal, rather than in a comparison of separately estimated models.

The practical implication is to specify exposure measurement and regularization together. When only the factor representation changes, transporting risk adjustments consistently preserves the intended stock-level penalty. Deliberate changes in the penalty or the mean inputs can then be evaluated on common strategy returns. This separates the information supplied by a factor model from the exposure preferences through which that information becomes a portfolio.

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Internet Appendix

Characteristic Geometry and Portfolio Choice

The main paper states the model, empirical comparisons, and their interpretation. This appendix supplies the ideal and implemented estimation algebra, benchmark specifications and estimation timing, portfolio and cost formulas, resampling definitions, and supplementary archived evidence. The source-to-result crosswalk and execution logs are supplied with the replication package.

IA.1. Estimation Algebra and the Recorded Implementation

IA.1.1. The ideal least-squares reduction

Within a fixed formation window, write $h_{s+1} = \Gamma x_{s+1}$, where $\Gamma = [\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha]$ and $x_{s+1} = (f'_{s+1}, g'_{s+1}, 1)'$. Omit the final column and regressor when the intercept is excluded. Define $m_s = X_s - W_s h_{s+1}$. Since X_s belongs to the range of W_s , the Moore–Penrose identities give

$$\begin{aligned} m'_s W_s^+ m_s &= X'_s W_s^+ X_s - 2X'_s h_{s+1} + h'_{s+1} W_s h_{s+1}, \\ \frac{\|r_{s+1} - Z_s h_{s+1}\|^2}{N_s} &= \frac{r'_{s+1} r_{s+1}}{N_s} - 2X'_s h_{s+1} + h'_{s+1} W_s h_{s+1}. \end{aligned} \quad (\text{IA1})$$

The two criteria differ only in their constant term. Multiplication by N_s therefore yields the equally weighted stock-month objective. Equal weighting of the normalized moment criterion instead yields equal-date average squared errors. These are different date-weight policies.

Unique factor profiling requires $\Gamma'_\beta W_s \Gamma_\beta \succ 0$, not full rank of the entire W_s . With loadings held fixed, the exact factor normal equation is

$$(\Gamma'_\beta W_s \Gamma_\beta) f_{s+1} = \Gamma'_\beta [X_s - W_s (\Gamma_\delta g_{s+1} + \Gamma_\alpha)]. \quad (\text{IA2})$$

With factors fixed, differentiating the objective in Γ gives

$$\left\{ \sum_s N_s (x_{s+1} x'_{s+1} \otimes W_s) \right\} \text{vec}(\Gamma) = \sum_s N_s \text{vec}(X_s x'_{s+1}). \quad (\text{IA3})$$

Here vec stacks columns. The identity $\text{vec}(W\Gamma xx') = (xx' \otimes W) \text{vec}(\Gamma)$ explains the Kronecker system. These are unconstrained conditional normal equations, not the first-order conditions of the fully constrained problem.

IA.1.2. The production moment ridge and block-solve jitters

The recorded production configurations replace W_s by $W_s^{\text{imp}} = W_s + \rho_W I$, with $\rho_W = 10^{-8}$. Expanding the inverse-weighted criterion using this matrix yields

$$\begin{aligned} & \sum_s N_s (X_s - W_s^{\text{imp}} h_{s+1})' (W_s^{\text{imp}})^{-1} (X_s - W_s^{\text{imp}} h_{s+1}) \\ &= \sum_s \|r_{s+1} - Z_s h_{s+1}\|^2 + \rho_W \sum_s N_s \|h_{s+1}\|^2 + C^{\text{imp}}, \end{aligned} \quad (\text{IA4})$$

where C^{imp} is parameter-free. Thus production perturbs the ideal criterion by an index penalty. Its invariance under a matched section map follows from preservation of h_{s+1} , not from an assertion that a small coefficient has no effect.

Separate jitters may be added to the linear systems in (IA2)–(IA3). The statistical block-loading penalty is set to zero in the frozen options. The moment ridge, block-solve jitters, factor-covariance ridge λ_F , and stock-space ridge λ_Σ are distinct operations. A complete convergence audit must use their actual configured values and the corresponding criterion.

IA.1.3. Normalization, budget enforcement, and stopping

The equalities in (2) are restored by matched transformations of the loading blocks and factor path, as in Proposition 1. Cholesky normalization, for example, uses $R'R = \Gamma'_\beta Q \Gamma_\beta$ and replaces Γ_β by $\Gamma_\beta R^{-1}$ and the latent factors by Rf . Removing a retained loading component in the latent span requires adding the compensating observed-factor or constant term to f . Both operations preserve fitted indices.

The intercept projection also weakly reduces its exposure. Writing $c = \Gamma'_\beta Q \Gamma_\alpha$ and using $\Gamma'_\beta Q \Gamma_\beta = I_K$ gives

$$(\Gamma_\alpha - \Gamma_\beta c)' Q (\Gamma_\alpha - \Gamma_\beta c) = \Gamma'_\alpha Q \Gamma_\alpha - c' c. \quad (\text{IA5})$$

Thus a second budget rescale is redundant after this projection in exact Q geometry. This identity need not hold if a fallback projects using one metric but measures feasibility using another; the same numerical budget then need not describe the same restriction.

By contrast, radial rescaling of an intercept that violates $\Gamma'_\alpha Q \Gamma_\alpha \leq \delta_\alpha$ is not generally fit-preserving. Nor is an unconstrained least-squares update followed by projection necessarily the exact constrained minimizer. Terminal slackness of the budget does not prove that an altered budget would leave the entire numerical path unchanged.

The documented iteration monitor is change in the unweighted managed-moment reconstruction norm, rather than the weighted objective in (7). All 3,090 recorded Core conditional-model windows specify a tolerance of 10^{-6} and a cap of 1,000 cycles. These dispatch settings override shorter defaults in the inherited toolkit options. Previous-window warm starts are used in the inspected NA-IPCA production driver. The common-initialization diagnostic instead uses a common singular-vector start and has one 1999 window reaching its cap. These runs must not be described as the same initialization experiment. Table IA.I reports the separate common-start diagnostic, including the capped window rather than replacing it.

Table IA.I. Common-initialization diagnostic, EWMA QZ-IPCA versus IPCA. The separate run uses 618 Core windows, a common singular-vector start, zero block-loading penalty, moment ridge 10^{-8} , tolerance 10^{-6} , and at most 1,000 iterations. Both lines meet their recorded stopping criterion in 617 windows; both reach the cap in June 1999. All 618 windows remain in the diagnostics below. Both weight comparisons hold the production IPCA residual diagonal common. The last row removes the factor-coordinate ridge from the re-estimated factor covariances, retaining the asset-space ridge and numerical eigenvalue floor. These comparisons concern spans and weights, not a verified constrained score condition or global optimum.

Diagnostic	Median	Maximum
Largest latent-span angle (degrees)	4.3×10^{-4}	1.1×10^{-2}
Relative forecast gap	4.7×10^{-6}	1.4×10^{-4}
Weight cosine, factor ridge retained	0.830	0.988
1-cosine, factor ridge removed	5.8×10^{-12}	4.6×10^{-9}

An audit of all 618 saved NA-IPCA endpoints confirms terminal feasibility: maximum exposure is 1.51×10^{-4} against the budget 0.035, maximum latent-normalization residual is 1.74×10^{-12} , and maximum cross-block residual is 5.27×10^{-15} . Nevertheless, all 618 records carry a convergence flag, including one at the iteration cap. In 404 windows the recorded monitor increases at least

once by more than 10^{-10} ; this is not a test of descent in the differently weighted objective. None of these records contains a budget-enforcement or fallback event log. Thus terminal slackness cannot establish that such events never occurred.

Execution success, a small reconstruction change, and a stationary constrained solution are different diagnostics. The companion theory's exactly profiled local estimator requires the last of these, together with appropriate localization. We do not infer that requirement from the archived termination flag. The release retains original sources and configuration provenance so that a future raw-panel rerun can record weighted-objective changes, score and constraint residuals, conditioning, and all budget-enforcement events.

IA.1.4. Preprocessing and information dates

The inspected production driver consumes a processed rank-based input, replaces nonfinite features and returns by zero, clips feature entries, and then applies training-window standardization. Thus it is not a complete-case procedure, and the clipping order differs from a standardize-then-clip pipeline. In the supplied processed Core panel, none of the 132 characteristic columns or the return column contains nonfinite values, and no feature lies outside the clipping threshold. The audit covers 1,362,000 Core stock-months, of which 1,244,002 are in the evaluation sample. It does not reconstruct preprocessing performed before the snapshot was created.

Four evaluation excess returns are below -1 . Such observations need not be invalid because the recorded return is net of a risk-free rate. Adding the locally stored next-month FF5 risk-free rate leaves one implied total return slightly below -1 (by 1.83 basis points). This flagged observation is retained, not silently clipped or deleted; reconciliation requires the original excess-return and delisting conventions.

The 120-month estimation window and the trailing metric must use only information available at portfolio formation. Training moments legitimately contain their paired realized returns; the next-month forecast cannot. Metric fitting-range checks are informative but do not independently verify characteristic reporting lags, return alignment, stock eligibility, or delisting treatment. Across 4,944 saved configuration records for eight Core model/metric sequences, every recorded estimation span is 120 months, the information cutoff equals the training end, and the evaluation dates run

consecutively from June 1973 to November 2024. These are configuration-record checks; they do not resolve the underlying input-date and selection questions.

IA.2. Benchmark Estimation and Portfolio Inputs

The benchmark comparisons in Section II.A vary the factor specification while retaining the portfolio rule. This section identifies what each benchmark estimates and what it supplies to that rule. Throughout, t is the portfolio-formation month, $\mathcal{W}_t = \{t - 120, \dots, t - 1\}$ is its training window, and $n = |\mathcal{W}_t| = 120$. Characteristics use the current window's training-sample transformation, as in the main text.

IA.2.1. The characteristic-instrumented FF5 benchmark

The observed-only specification uses the market excess return, size, value, profitability, and investment factors of Fama and French (2015), collected in $g_{s+1} \in \mathbb{R}^5$. Setting the latent and intercept blocks to zero in (1) gives

$$r_{i,s+1} = z'_{i,s} \Gamma_{\delta,t} g_{s+1} + \varepsilon_{i,s+1}, \quad \beta_{i,s}^{\text{FF5}} = \Gamma'_{\delta,t} z_{i,s}. \quad (\text{IA6})$$

Here $\Gamma_{\delta,t} \in \mathbb{R}^{L \times 5}$ is a common loading map and $\beta_{i,s}^{\text{FF5}}$ is the stock's five-dimensional exposure. With $L = 132$, the map contains 660 coefficients. Exposures vary with characteristics even though the map is fixed within each window.

Before the moment ridge described in Internet Appendix IA.1, the loading estimate solves the pooled least-squares problem

$$\hat{\Gamma}_{\delta,t} \in \arg \min_{G \in \mathbb{R}^{L \times 5}} \sum_{s \in \mathcal{W}_t} \|r_{s+1} - Z_s G g_{s+1}\|^2. \quad (\text{IA7})$$

Each regressor is a characteristic multiplied by a realized observed factor. The criterion weights stock-month observations equally. The loading normal equation is (IA3) with $x_{s+1} = g_{s+1}$; the recorded implementation uses W_s^{imp} in that system. With no latent factors, there is no factor-update

step. This estimates how characteristics map into observed-factor exposures, rather than estimating unrestricted betas stock by stock or performing a Fama–MacBeth second-pass regression.

At formation, the stock-exposure matrix and expected-return input are

$$B_t^{\text{FF5}} = Z_t \widehat{\Gamma}_{\delta,t}, \quad \widehat{m}_{g,t} = \frac{1}{n} \sum_{s \in \mathcal{W}_t} g_{s+1}, \quad \widehat{\mu}_t^{\text{FF5}} = B_t^{\text{FF5}} \widehat{m}_{g,t}. \quad (\text{IA8})$$

The factor covariance is the centered sample covariance of this same observed-factor history. Thus the FF5 row evaluates a characteristic-based exposure model using only the five observed factors. Relative to the mixed models, it is an observed-factor reference, not an eight-factor, dimension-matched specification.

IA.2.2. RP-PCA on characteristic-managed returns

For the statistical benchmarks, stack the managed returns $X_s = Z'_s r_{s+1} / N_s$ into $M_t \in \mathbb{R}^{n \times L}$, whose row for training month s is X'_s . The input is therefore a balanced panel of characteristic-managed returns, not a balanced panel of individual stock returns. Define its mean and centered matrix by

$$\bar{x}_t = \frac{1}{n} M'_t \mathbf{1}_n, \quad M_{c,t} = M_t - \mathbf{1}_n \bar{x}'_t,$$

where $\mathbf{1}_n$ is the n -vector of ones. Under the recorded normalization settings, the RP-PCA estimator of Lettau and Pelger (2020) selects the leading eight eigenvectors of

$$\begin{aligned} S_{\gamma,t} &= \frac{1}{n} M'_t \left(I_n + \frac{\gamma_{\text{RP}}}{n} \mathbf{1}_n \mathbf{1}'_n \right) M_t \\ &= \frac{1}{n} M'_{c,t} M_{c,t} + (1 + \gamma_{\text{RP}}) \bar{x}_t \bar{x}'_t, \quad \gamma_{\text{RP}} = 3. \end{aligned} \quad (\text{IA9})$$

The equality follows by decomposing M_t into its centered part and mean. The additional mean term gives expected-return information a role in selecting directions, alongside covariance variation. This extraction matrix weights training months equally, whereas the FF5 loss aggregates stock-level squared errors across the unbalanced panel.

Let $V_t \in \mathbb{R}^{L \times 8}$ collect these orthonormal eigenvectors, with their signs oriented by the training

factor means. The production call applies no additional managed-return standardization, factor-variance normalization, or factor orthogonalization.⁵ Its returned loading matrix and factor history consequently reduce to

$$\widehat{\Gamma}_{\beta,t}^{\text{RP}} = V_t, \quad \widehat{F}_t^{\text{RP}} = M_t V_t, \quad \widehat{m}_{f,t}^{\text{RP}} = \frac{1}{n} (\widehat{F}_t^{\text{RP}})' \mathbf{1}_n = V_t' \bar{x}_t. \quad (\text{IA10})$$

Factor histories use uncentered managed returns. The portfolio adapter forms $B_t^{\text{RP}} = Z_t V_t$ and $\widehat{\mu}_t^{\text{RP}} = B_t^{\text{RP}} \widehat{m}_{f,t}^{\text{RP}}$, and passes the centered sample covariance of $\widehat{F}_t^{\text{RP}}$ to the common risk construction. The toolkit's auxiliary regression intercepts are not passed to this portfolio as a retained mean block. This separates the RP-PCA extraction criterion from the NA-IPCA intercept-input exercise.

The mean convention also explains the excluded PCA configuration. It selects covariance eigenvectors V_t^{PCA} but returns the centered history $M_{c,t} V_t^{\text{PCA}}$, whose sample mean is zero. With no separate mean block, the portfolio mean is zero in exact arithmetic. Section II.C records the resulting numerical degeneracy. This benchmark definition matters for interpreting Table VIII: RP-PCA supplies a mean-sensitive eight-factor alternative, whereas the archived centered-PCA mean rule does not supply a usable portfolio comparison.

IA.3. Estimation Timing and the Strict-24 Sample

The design uses two histories for different tasks. A 120-month window estimates the current loading maps and factor-return history. Separately, up to 24 previously saved intercept-loading estimates determine the loading-instability adjustment. Figure IA.1 places both inputs relative to portfolio formation and the next-month payoff.

IA.3.1. Information used at formation

At month t , training uses the pairs $\{Z_s, r_{s+1}, g_{s+1}\}_{s \in \mathcal{W}_t}$. The last training pair is (Z_{t-1}, r_t, g_t) ; the current cross section Z_t is used to forecast r_{t+1} . The fit provides one current loading map and 120 dates of factor realizations. Denote the fitted latent factor at training date $s + 1$ by $\widehat{f}_{s+1}^{(t)}$, where the superscript identifies the formation-window fit, and set $\widehat{F}_{s+1}^{(t)} = (\widehat{f}_{s+1}^{(t)}, g'_{s+1})'$, omitting absent

⁵The production options `stdnorm`, `variancennormalization`, and `orthogonalization` are each set to zero.

blocks for the relevant model. The factor mean and covariance inputs are

$$\begin{aligned}\widehat{m}_{F,t} &= \frac{1}{120} \sum_{s \in \mathcal{W}_t} \widehat{F}_{s+1}^{(t)}, \\ \widehat{C}_{F,t} &= \frac{1}{119} \sum_{s \in \mathcal{W}_t} (\widehat{F}_{s+1}^{(t)} - \widehat{m}_{F,t})(\widehat{F}_{s+1}^{(t)} - \widehat{m}_{F,t})'.\end{aligned}\tag{IA11}$$

These are moments of monthly factor returns within the current fit. They are distinct from the covariance of past loading estimates.

For the intercept-inclusive rule, the current estimate $\widehat{\Gamma}_{\alpha,t}$ enters the mean forecast. The risk adjustment instead reads $\widehat{\Gamma}_{\alpha,t-m_t}, \dots, \widehat{\Gamma}_{\alpha,t-1}$, where m_t is the number of available prior monthly fits, capped at 24. It excludes the current estimate. Equation (IA14) gives the resulting loading covariance; equation (10) maps the portfolio weights formed at t into the realized payoff at $t + 1$.

IA.3.2. Why strict-24 starts at the twenty-fifth fit

The first recorded rolling fit, dated June 1973, already has its 120-month training sample but no prior saved loading estimates. The twenty-fifth fit, dated June 1975, is the first with 24 predecessors: June 1973 through May 1975. Its training characteristics run from June 1965 through May 1975, paired with returns and observed factors from July 1965 through June 1975; the portfolio payoff is realized in July 1975.

The available-history sample retains 618 formation months from June 1973 through November 2024. Restricting evaluation to a full prior loading history removes the first 24 months and leaves 594 formation months, June 1975 through November 2024. It changes the evaluation sample, not the 120-month estimation window. Benchmarks without an intercept adjustment are evaluated on these same dates in a strict-24 comparison.

Strict-24 thus holds the historical input length fixed when comparing portfolio rules. Adjacent fits share 119 training months, so it does not provide 24 independent loading estimates. Internet Appendix IA.4.3 discusses the interpretation and rank of the loading-instability input; Internet Appendix IA.1.4 states the remaining upstream information-date checks.

A. Two histories feed one formation-date portfolio

Current model fit

120 pairs: (Z_s, r_{s+1}, g_{s+1})

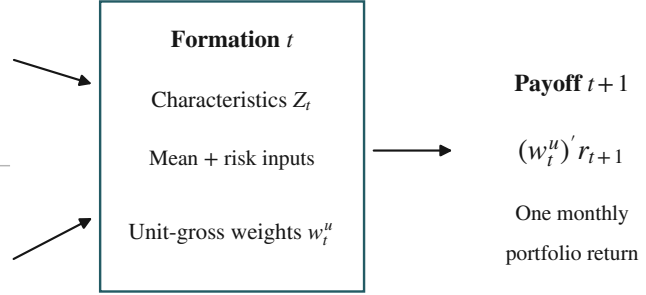
$s = t - 120, \dots, t - 1$; last payoff r_t

Current loadings; factor mean and covariance

Prior loading estimates

24 vectors: $\hat{\Gamma}_{a,t-24}, \dots, \hat{\Gamma}_{a,t-1}$

Instability input; current estimate excluded



B. June 1975: first formation with 24 prior estimates

Training characteristics	June 1965 - May 1975	120 months
Paired returns and factors	July 1965 - June 1975	120 months
Prior intercept-loading estimates	June 1973 - May 1975	24 vectors
Formation / next-month payoff	June 1975 / July 1975	1 portfolio return

Figure IA.1. Training, historical loading estimates, and portfolio formation. Panel A distinguishes the 120 training pairs from the 24 prior monthly loading estimates used by the strict-24 intercept-instability rule. Both feed the formation-date portfolio; its return is realized one month later. Panel B gives the first strict-24 observation under the paper's formation-date convention. The current loading is used in the mean forecast but is excluded from that month's historical-loading covariance. The diagram describes the implemented indexing rule; it is not an independent verification of the source data's publication dates.

IA.4. Portfolio Solves, Instability Inputs, and Trading Costs

IA.4.1. From an intercept budget to portfolio exposure

For $Q \succ 0$ and any given stock-weight vector $w \in \mathbb{R}^{N_t}$, the portfolio's fitted intercept contribution is $w'Z_t\Gamma_\alpha$. Writing it as an inner product and applying Cauchy–Schwarz gives

$$\begin{aligned} |w'Z_t\Gamma_\alpha| &= |(Q^{-1/2}Z_t'w)'(Q^{1/2}\Gamma_\alpha)| \\ &\leq (w'Z_tQ^{-1}Z_t'w)^{1/2}(\Gamma_\alpha'Q\Gamma_\alpha)^{1/2} \\ &\leq \sqrt{\delta_\alpha} (w'Z_tQ^{-1}Z_t'w)^{1/2}. \end{aligned} \tag{IA12}$$

The last inequality uses (5). It makes the economic role of the budget precise: the allowed intercept contribution depends on how much characteristic exposure a portfolio takes. The matrix $Z_tQ^{-1}Z_t'$ measures exposure in the dual characteristic geometry; it is not a return-covariance matrix. Consequently, this bound controls a mean component relative to exposure, not relative to payoff risk. An interpretation as a Sharpe-ratio or stochastic-discount-factor distance bound would require additional pricing and covariance restrictions.

IA.4.2. The low-rank solve

Suppose the stock-space risk matrix is $D + VCV'$, with $D \succ 0$ diagonal and $C \succeq 0$. This encompasses factor and intercept-instability terms after including $\lambda_\Sigma I$ in D . Choose the symmetric square root of C and put $U = VC^{1/2}$. The Woodbury identity gives

$$(D + UU')^{-1}\mu = D^{-1}\mu - D^{-1}U(I + U'D^{-1}U)^{-1}U'D^{-1}\mu. \tag{IA13}$$

Only the smaller middle system must be inverted. The formula also permits singular C , because it does not invert C itself. It is algebraically the same portfolio solve, subject to numerical precision; it does not create a different risk model.

The recorded $\widehat{D}_t$ is based on stock residuals after subtracting factor contributions. In the intercept-inclusive specification this is not the same as subtracting the complete fitted mean. Consequently, recomputing this diagonal after a metric-induced intercept transfer need not preserve

it. The nine-convention sensitivity exercise holds the production diagonal fixed and should be read as an isolated risk-input comparison.

IA.4.3. What the two intercept risk inputs measure

For m_t prior intercept estimates, the trailing input is

$$V_{\Gamma_{\alpha,t}} = \frac{1}{m_t - 1} \sum_{j=1}^{m_t} (\widehat{\Gamma}_{\alpha,t-j} - \bar{\Gamma}_{\alpha,t})(\widehat{\Gamma}_{\alpha,t-j} - \bar{\Gamma}_{\alpha,t})', \quad (\text{IA14})$$

for $m_t \geq 2$, where $\bar{\Gamma}_{\alpha,t}$ is their average. The rule uses the most recent available estimates, up to 24, and sets $V_{\Gamma_{\alpha,t}} = 0$ when $m_t < 2$. Thus $m_t = 24$ throughout the strict-24 sample. For $m_t \geq 2$, its rank is at most $m_t - 1$. Adding $Z_t V_{\Gamma_{\alpha,t}} Z_t'$ penalizes positions exposed to instability in the recorded coefficient directions.

Those coefficients may be expressed in different within-window standardizations. Without transporting them to a common feature coordinate system, their dispersion combines coefficient movement with changes in preprocessing. This is another reason to describe the matrix as a numerical loading-instability proxy rather than a calibrated sampling covariance.

The alternative full-rank matrix in Table IA.V uses a within-window, residual-based construction. Its equal-date weighting differs from the production stock-month weighting; its residuals omit the intercept; and it does not differentiate the section restriction $\Gamma'_\beta Q \Gamma_\alpha = 0$. Conditional on fixed full-rank latent loadings, that restriction leaves only $L - K$ admissible intercept directions. A full-rank L -dimensional matrix is therefore not automatically the covariance of that conditional constrained estimator. The experiment is retained as a sensitivity calculation, not as a validation of complete estimation-risk inference.

IA.4.4. The inherited cost proxy

Let $\text{TO}_t^{\text{raw}} = \frac{1}{2} \|w_t - w_{t-1}\|_1$ denote the saved one-way raw turnover, aligning stock identifiers and setting absent positions to zero, and $q_t^{\text{proxy}} = \text{TO}_t^{\text{raw}} / G_t$. The archived proportional-cost series is

$$r_{t+1}^{u,\text{proxy}}(c) = r_{t+1}^u - \frac{c}{10^4} q_t^{\text{proxy}}, \quad (\text{IA15})$$

for c basis points. Even ignoring portfolio drift,

$$\frac{\|w_t - w_{t-1}\|_1}{2G_t} \neq \frac{1}{2} \left\| \frac{w_t}{G_t} - \frac{w_{t-1}}{G_{t-1}} \right\|_1 \quad (\text{IA16})$$

in general. Actual trades compare current targets with return-drifted pre-trade holdings on a common stock identifier set, including cash and financing conventions. The saved scalar proxy does not supply those holdings.

Costs can change volatility as well as the mean. Bilinearity of covariance gives, with $k = c/10^4$,

$$\text{Var}(r^u - kq) = \text{Var}(r^u) + k^2 \text{Var}(q) - 2k \text{Cov}(r^u, q). \quad (\text{IA17})$$

For positive mean return and positive mean proxy turnover, the zero-mean cost threshold is $c^* = 10^4 \bar{r}^u / \bar{q}$. This follows from the mean, not from constant net volatility. It is a threshold under the proxy, not a verified break-even transaction cost of the normalized strategy.

For historical reconciliation, the proxy calculation gives 50-basis-point Sharpe ratios of 0.27 and 0.57 for IPCA and rolling-mean QZ-IPCA on the 618-month sample. On the strict-24 sample it gives 1.62 for complete IAB. These figures are retained in the replication outputs with a proxy label. They are not evidence of achievable net returns; the main tables report gross performance until holdings-based costs are available.

Figure IA.2 applies the same proxy on the common 594-month sample to complete IAB, RP-PCA, and all three QZ-IPCA constructions. The full series is recalculated at every charge, including its volatility. Complete IAB has the higher gross Sharpe, but its zero-mean proxy threshold is approximately 136 basis points, compared with 142 for RP-PCA and 111–115 for QZ-IPCA. Thus, a higher gross Sharpe does not mechanically imply a higher proxy cost threshold. The comparison identifies which conclusions are sensitive to the archived turnover measure and where a holdings-based implementation audit is most valuable.

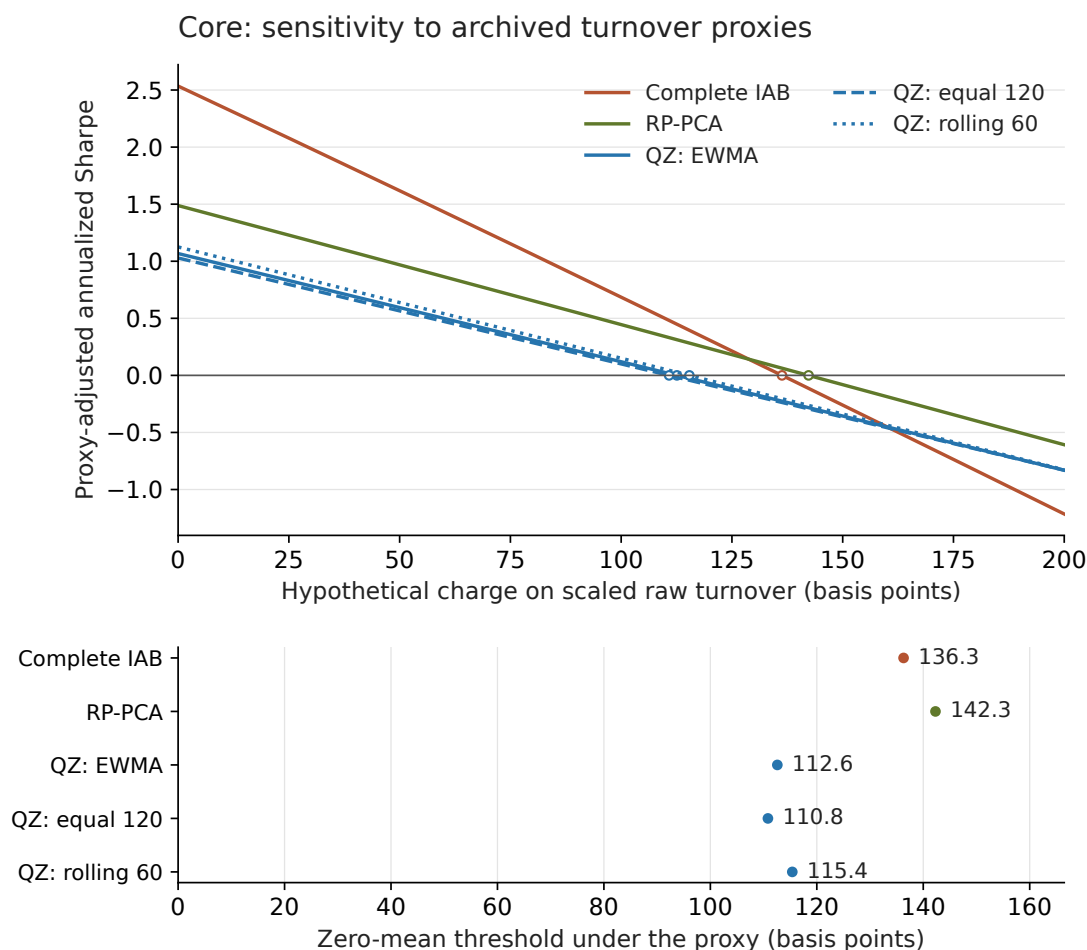


Figure IA.2. Sensitivity to the archived turnover proxy, Core, June 1975–November 2024 (594 formation months). The upper panel plots the annualized Sharpe ratio of $r_{t+1}^u - (c/10^4)q_t^{\text{proxy}}$ for each hypothetical charge c . The lower panel reports $c^* = 10^4 \bar{r}^u / \bar{q}^{\text{proxy}}$ calculated from the same observations; the open circles above locate these zero-mean thresholds. Each curve's standard deviation is recomputed after the deduction. The proxy is saved raw turnover divided by contemporaneous raw gross exposure, not turnover reconstructed from normalized, return-drifted holdings. These are sensitivity calculations, not verified net Sharpe ratios or implementable break-even trading costs.

IA.5. Metric Definitions and Paired Time-Series Inference

IA.5.1. The recorded metric averages

Number the 120 training months from oldest to newest by $j = 0, \dots, 119$. Thus $s = t - 120 + j$ and $W_{t,j}^{\text{imp}} = W_s^{\text{imp}}$. Writing their implementation Gram matrices in this indexing, the EWMA construction is

$$\widehat{Q}_t^{\text{EWMA}} = \sum_{j=0}^{119} \frac{d^{119-j}}{\sum_{k=0}^{119} d^k} W_{t,j}^{\text{imp}}, \quad d = 2^{-1/60}. \quad (\text{IA18})$$

The alternatives average either all 120 matrices or only $j = 60, \dots, 119$. The archived code calls these `raw_mean_w` and `rolling_mean`, respectively. All three use equal-date Gram inputs before their stated time weighting, not an additional stock-count weight. The production construction forms an augmented constant-and-feature Gram matrix and passes its 132-dimensional feature block to the estimator.

IA.5.2. Forecast and weight diagnostics

For the common stock set at formation date t , define the relative forecast gap and cross-model weight cosine by

$$d_{\mu,t} = \frac{\|\widehat{\mu}_{Q,t} - \widehat{\mu}_{I,t}\|_2}{\|\widehat{\mu}_{I,t}\|_2}, \quad c_{w,t} = \frac{w'_{Q,t} w_{I,t}}{\|w_{Q,t}\|_2 \|w_{I,t}\|_2}. \quad (\text{IA19})$$

These require nonzero denominators. The absolute forecast diagnostic is $\max_t \|\widehat{\mu}_{Q,t} - \widehat{\mu}_{I,t}\|_\infty$ in monthly return units. The relative weight gap replaces the forecast vectors in $d_{\mu,t}$ by raw weights. It tests scale as well as direction, whereas the cosine does not.

IA.5.3. Paired bootstrap statistics and protocols

Moving-block resampling draws consecutive observations rather than isolated months, following the dependent-data bootstrap principle of Künsch (1989). The recorded implementation uses blocks of 12 months, concatenates sampled blocks to the evaluation length, and applies the same indices to both strategies. Gross exposure and turnover are carried jointly when proxy-cost statistics are recomputed.

For the 618-month rolling-metric pair, the mean-return test uses 10,000 draws and seed 20260713. With $d_{t+1} = r_{Q,t+1}^u - r_{I,t+1}^u$, percentile intervals are calculated from the resampled means. The two-sided null calculation resamples the centered series $d_{t+1} - \bar{d}$ and compares its mean magnitude with $|\bar{d}|$. The recorded risk-statistic bootstrap uses 10,000 draws and seed 20260714 and recomputes each strategy's Sharpe ratio before taking their difference:

$$\Delta \text{SR}^{(b)} = \sqrt{12} \left\{ \frac{\bar{r}_Q^{u,(b)}}{s_Q^{u,(b)}} - \frac{\bar{r}_I^{u,(b)}}{s_I^{u,(b)}} \right\}. \quad (\text{IA20})$$

This is not the Sharpe ratio of the return-difference series.

For the complete-versus-off comparisons in Table VI, the paired mean tests use the recorded 12-month, 10,000-draw specification with seed 20260714. Reported p -values use the finite Monte Carlo correction $(1 + N_{\text{extreme}})/(10,000 + 1)$. Holm adjustment is applied over the four universes within each reported sample. It does not adjust for all metric, multiplier, model, or subperiod choices in the broader research program.

For Table VII, percentile intervals use seed 20260714, whereas recentered mean tests use seed 20260713. These pointwise tests are not the capitalization-wise tests above and do not apply Holm adjustment.

A fixed 12-month block retains within-block dependence, not every possible dependence induced by overlapping 120-month estimation windows or structural change. Valid asymptotic block-bootstrap arguments require conditions on the return process and a suitable block-length sequence. The finite design is a stated approximation, not a theorem supplied by the companion loading-inference result. We therefore add 24- and 60-month blocks as retrospective sensitivity checks, keeping all comparisons and the seed fixed rather than selecting a block length by significance.

IA.5.4. Provenance and reporting precision

The archived EWMA forecast and risk-matrix diagnostics cover 618 months. The corrected reconstruction uses saved factors in all 618 windows of both models: observed FF5 factors must be dated one month after the characteristics paired with them, as in the production factor history.

The original 618-month QZ-IPCA performance interval uses the rolling mean. On that same available-history sample, the archived EWMA and 120-month equal-weight Sharpe ratios are 0.91 and 0.85. On the strict-24 sample the corresponding values are 1.07 and 1.03, versus 1.13 for the rolling mean. These labels must travel with the series in any subsequent analysis.

Every reported statistic should be computed from unrounded monthly data and rounded once. In particular, the 0.37 Sharpe difference comes from an unrounded value near 0.368; subtracting the displayed 1.00 and 0.64 instead gives 0.36. Different recorded random seeds can also give slightly different Monte Carlo endpoints for the same mean statistic. The release preserves both protocols rather than silently merging them.

This revision applies the same seeds and block design to all three recorded metrics and both common evaluation samples. The primary 618-month EWMA Sharpe-difference interval is $[-0.02, 0.58]$. Table IA.II provides the strict-24 results alongside monthly mean differences. These are retrospective sample-sensitivity results, not predeclared tests, and do not correct for specification search.

Table IA.II. Paired metric comparison on the strict-24 sample. Differences are QZ-IPCA minus IPCA on 594 common Core formation months, June 1975–November 2024, using unit-gross returns before costs. The 95% intervals are pointwise paired percentile intervals with 12-month blocks, 10,000 draws, and seed 20260714. The mean difference is in basis points per month. These retrospective sample-sensitivity comparisons do not adjust for specification selection; the longer-block metric checks in Table IA.VI use 618 months instead.

Metric	Δ Sharpe	95% interval	Δ mean (bp)	95% interval
EWMA	0.38	[0.09, 0.64]	6.19	$[-4.12, 17.68]$
Equal 120	0.34	[0.08, 0.57]	5.57	$[-4.41, 16.48]$
Rolling 60	0.44	[0.11, 0.75]	7.18	$[-3.82, 19.76]$

IA.6. Measuring the Characteristic Geometry

IA.6.1. Rolling production-metric diagnostics

The main-text diagnostic uses the exact saved 132×132 feature-block Gram matrix supplied to each EWMA QZ-IPCA fit. It does not use the augmented intercept row or the separate public-panel illustration below. The recorded training endpoint precedes formation in all 618 windows;

this verifies the saved date convention, not the publication lag of every underlying input. The replication module preserves the matrices and their source hashes and reconciles their condition numbers with the recorded metadata.

Let $\ell_{1,t} \geq \dots \geq \ell_{L,t} > 0$ be the eigenvalues of $\widehat{Q}_t$, with $L = 132$, and set $p_{j,t} = \ell_{j,t} / \sum_k \ell_{k,t}$. The spectral participation dimension and adjacent-window spectral drift are

$$d_t^{\text{part}} = \frac{1}{\sum_{j=1}^L p_{j,t}^2} = \frac{(\text{tr } \widehat{Q}_t)^2}{\text{tr}(\widehat{Q}_t^2)}, \quad v_t^{\text{spec}} = \frac{1}{2} \sum_{j=1}^L |p_{j,t} - p_{j,t-1}|. \quad (\text{IA21})$$

The trace identity follows from orthogonal diagonalization. Cauchy–Schwarz gives $1 \leq d_t^{\text{part}} \leq L$; equality at L means equal eigenvalues. The drift compares distributions of second-moment mass across ordered spectral positions and lies in $[0, 1]$. It requires two windows, so the first formation month has no drift value. Both statistics are unchanged by common positive scaling and coordinate permutations. They are not invariant to arbitrary feature-specific rescaling, and the drift does not measure changes in eigenspaces.

Table IA.III. The recorded rolling characteristic geometry. The sample contains all 618 saved EWMA feature-block Gram matrices, June 1973–November 2024, with 132 characteristics throughout. The participation dimension summarizes the dispersion of eigenvalue shares; it is not the number of priced factors. Spectral drift is defined in (IA21) and has 617 observations. The condition number uses the largest and smallest eigenvalues.

Diagnostic	Minimum	Median	Maximum
Participation dimension	17.36	19.83	22.63
Largest eigenvalue share (%)	11.76	14.24	15.21
Spectral drift (percentage points)	0.033	0.063	0.115
Condition number	1,735	5,922	13,747

Window-specific preprocessing hashes differ in all 617 transitions. Consequently, the paths include changing standardization, clipping, and stock samples as well as changing characteristic relationships. The diagnostics describe the geometry implemented by the estimator; they do not separately identify any of these sources. No crisis dates, state thresholds, or return-based tuning enter the calculation. All matrices remain positive definite, so concentration here is distinct from

rank failure. A test of the investment value of adaptation would additionally require a common-information, common-policy comparison with a frozen ruler.

IA.6.2. A separate public-panel illustration

The archived public-panel illustration is separate from the 132-feature estimation sample. It uses 153 characteristics over 1962–2025, with 768 monthly cross sections and 36,621 distinct firms. Characteristics are rank-scaled to $[-1/2, 1/2]$ within month; missing entries are assigned the center, and characteristics with fewer than ten observations are set to the center in that month. The numerical results below are retained from that measurement record, not reconstructed from the production panel in this revision.

The measured object is

$$\hat{Q}_{\text{public}} = \frac{1}{768} \sum_s \frac{Z'_s Z_s}{N_s}. \quad (\text{IA22})$$

Its eigenvalue shares describe concentration of squared characteristic exposure, not risk premia or systematic return variance. The archived condition number is approximately 2.3×10^3 , the largest eigenvalue accounts for approximately 13% of the trace, and the first five account for approximately 38%, as shown in Figure IA.3. This illustrates how marginal rank transformations can coexist with substantial overlap across characteristics.

Missing-value centering reduces a sparsely observed characteristic's second moment. Thus coverage as well as economic variation shapes the measured spectrum. The production-matrix calculation above supplies the corresponding check on the actual estimation sample. The two panels must not be pooled: they use different characteristics and preprocessing. Neither varies the characteristic count within an otherwise fixed investment experiment.

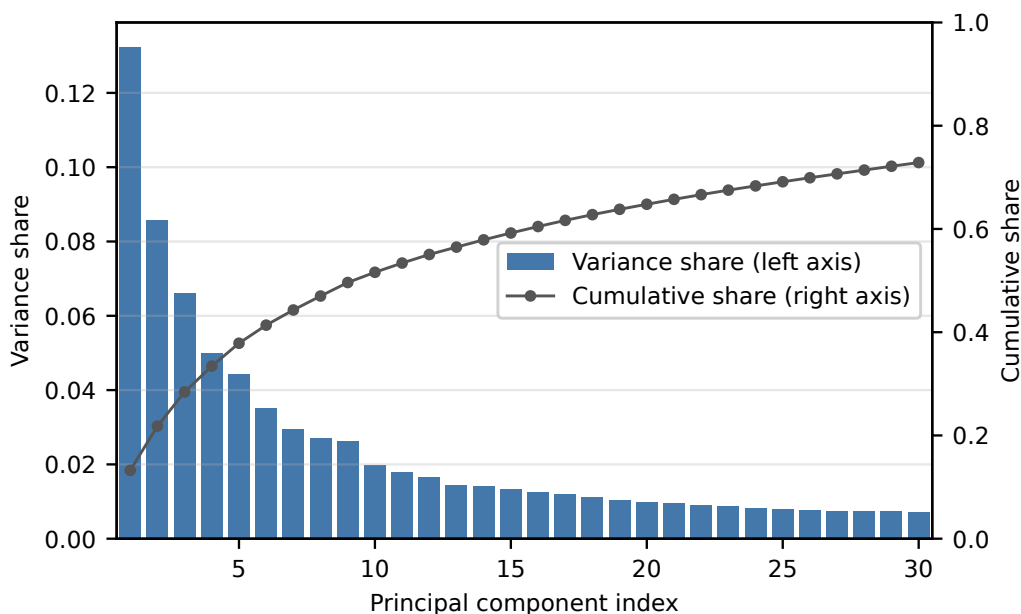


Figure IA.3. Archived public-panel characteristic spectrum. The panel contains 153 rank-transformed characteristics over 1962–2025, not the production 132-feature sample. Bars and the cumulative line report shares of the Gram matrix trace. The source graphic uses the term variance share; because the matrix is not generally demeaned, the precise interpretation is a share of second-moment mass. This supplementary graphic has not been regenerated from the public raw panel in this revision.

IA.7. Supplementary Gross-Return Evidence

The following exhibits use archived monthly portfolio records. They extend the gross-return comparisons in the main text without introducing another metric-selection or significance rule.

IA.7.1. Stock-level forecast evidence

Table IA.IV makes the prediction comparison in Section VI.B explicit. The forecast-evaluation archive defines its own common stock-month set; its 1,210,039 observations should not be confused with the 1,244,002 stock-month count for the longer 618-month production evaluation. A negative predictive R^2 means that the forecasts have larger pooled squared error than zero. Positive IC can coexist with this result because correlation does not assess forecast scale.

Table IA.IV. Forecast metrics on the common 594-month Core sample. The sample is June 1975–November 2024, with 1,210,039 stock-months in the saved forecast evaluation. IC is the mean monthly cross-sectional Pearson correlation of forecasts and excess returns; rank IC uses Spearman correlations. Predictive R^2 is $1 - \sum (r - \hat{r})^2 / \sum r^2$, pooled over stock-months, with zero as the benchmark forecast. All metrics use decimal units. QZ-IPCA uses rolling 60, not EWMA. The final row omits the NA-IPCA intercept and does not evaluate the complete IAB mean.

Forecast	IC	Rank IC	Predictive R^2
FF5	0.0159	0.0170	−0.0004
IPCA	0.0254	0.0249	−0.0188
QZ-IPCA, rolling 60	0.0254	0.0249	−0.0188
RP-PCA	0.0364	0.0415	−0.2568
NA-IPCA, intercept off	−0.0021	−0.0046	−0.0174

IA.7.2. Risk-input sensitivity beyond the primary ablation

Table IA.V preserves the archived 618-month multiplier sweep. It is separate from the same-sample 594-month ablation in the main text. With the intercept already in the mean, the trailing proxy changes the Sharpe ratio from 2.4686 at $\kappa_U = 0$ to 2.4926 at $\kappa_U = 1$. The alternative full-rank matrix gives 2.4758.

Table IA.V. Intercept-instability sensitivity, Core available-history sample. Annualized gross Sharpe ratios use 618 unit-gross monthly returns, June 1973–November 2024. Estimates and the mean input are fixed while κ_U varies. The trailing input uses up to 24 prior loading estimates; the alternative is the archived full-rank residual-based sensitivity matrix, not a validated covariance of the constrained loading estimator.

Portfolio input	Trailing proxy	Full-rank sensitivity
Intercept off, no instability term	0.32	0.32
Intercept in mean, $\kappa_U = 0$	2.4686	2.4686
Intercept in mean, $\kappa_U = 0.25$	2.4755	2.4705
Intercept in mean, $\kappa_U = 0.5$	2.4818	2.4723
Intercept in mean, $\kappa_U = 1$	2.4926	2.4758
Intercept in mean, $\kappa_U = 2$	2.5096	2.4825

The two $\kappa_U = 0$ entries agree by construction because the risk adjustment is absent. These fixed-input re-solves carry no sampling intervals. The beta-only label in the archived code refers to the intercept-off row, not to the mean-only $\kappa_U = 0$ row.

The alternative is a residual-based sensitivity matrix, not a validated sampling covariance for the production intercept estimator: its date weights differ, its residuals omit the fitted intercept, and

it does not differentiate the section constraints or joint factor estimation. Full rank alone does not remedy these differences. A correct covariance for a constrained estimator may itself be singular. The informative comparison is whether the matrix estimates uncertainty for the intended estimator, not whether every ambient direction receives positive variance.

The nine-convention diagnostic is also deliberately limited. It holds the production mean and systematic covariance fixed and re-prices only the trailing instability input. Its $\kappa_U = 1$ Sharpe ratios range from 2.4895 to 2.4938, measuring that input's sensitivity. The $\kappa_U = 0$ equality is imposed by design, and the calculation does not establish full-strategy invariance or feasibility under all nine metrics. The nine recorded labels include the baseline EWMA and an equivalent 60-month-half-life EWMA specification, not nine independent metric choices.

IA.7.3. Sensitivity to block length

Table IA.VI reports pointwise paired percentile intervals with 12-, 24-, and 60-month blocks, 10,000 draws, and seed 20260714. Longer blocks retain more within-block dependence but provide fewer resampled blocks; they are sensitivity choices, not an estimated optimal sequence. The rolling-60 metric's Sharpe interval includes zero at the two longer lengths. By contrast, the mean-only versus off comparison remains large and positive, and the complete versus mean-only Sharpe difference remains small and positive. Monthly mean intervals for the latter include zero at every length; the replication CSV provides both mean and Sharpe intervals for every row.

Table IA.VI. Paired Sharpe-difference intervals across block lengths. Entries are pointwise 95% percentile intervals from 10,000 paired draws with seed 20260714. Metric comparisons use the 618-month Core sample; input comparisons use the 594-month strict-24 sample. All returns use unit-gross exposure and are gross of costs. The three block lengths are reported jointly as retrospective sensitivity checks.

Comparison	12 months	24 months	60 months
EWMA minus IPCA	[−0.019, 0.577]	[−0.048, 0.591]	[−0.035, 0.639]
Equal 120 minus IPCA	[−0.060, 0.512]	[−0.089, 0.533]	[−0.061, 0.573]
Rolling 60 minus IPCA	[0.019, 0.682]	[−0.012, 0.693]	[−0.010, 0.757]
Mean-only minus off	[1.644, 2.679]	[1.582, 2.840]	[1.421, 3.183]
Complete minus mean-only	[0.010, 0.035]	[0.011, 0.036]	[0.012, 0.035]

IA.7.4. Raw solves and unit-gross portfolios

Table IA.VII compares gross exposure and Sharpe ratios before and after unit-gross normalization on the 594-month Core sample. The raw mean–variance scale omits an explicit risk-aversion multiplier. It should not be interpreted as uniquely required economic leverage. The unit-gross policy removes a common positive scaling of each month’s weights but can alter time-series performance when the scale varies over time. The IAB raw return series includes observations below -100% ; its raw Sharpe is consequently only an arithmetic moment ratio. Its mean raw gross exposure is approximately 501 and 83 raw monthly returns are below -100% . The recorded outputs do not isolate the cause of this scale; unit-gross normalization defines a different leverage policy, rather than validating the unnormalized positions.

Table IA.VII. Raw-solve and unit-gross reconciliation. Core strict-24 sample, June 1975–November 2024 (594 months). Gross exposure is measured under the saved raw-solve convention. Sharpe ratios use monthly arithmetic returns and are gross of costs. The IAB raw series contains returns below -100% and is not an implementable wealth path. Unit gross is the portfolio policy evaluated in the main text.

Specification	Mean gross	Raw Sharpe	Unit-gross Sharpe
FF5	1.98	0.77	0.75
RP-PCA	0.53	1.53	1.49
IPCA	3.79	0.49	0.69
QZ-IPCA, rolling mean	2.37	1.00	1.13
QZ-IPCA, EWMA	2.37	0.97	1.07
NA-IPCA, complete IAB	501	2.48	2.53
NA-IPCA, intercept-off ablation	2.27	0.38	0.39

IA.7.5. Subperiod comparison

The archived strict-24 record splits the 594 months into three consecutive 198-month blocks. The complete IAB gross Sharpe ratios are 3.97, 3.03, and 1.13. These differ from the three 206-month blocks used for the 618-month IPCA–QZ-IPCA comparison in the main text. The common deterioration in the last block is a statement about measured strategy returns, not a direct estimate of time-varying risk premia. Figure IA.4 instead uses overlapping 120-month windows to display the evolution of the complete-versus-off comparison.

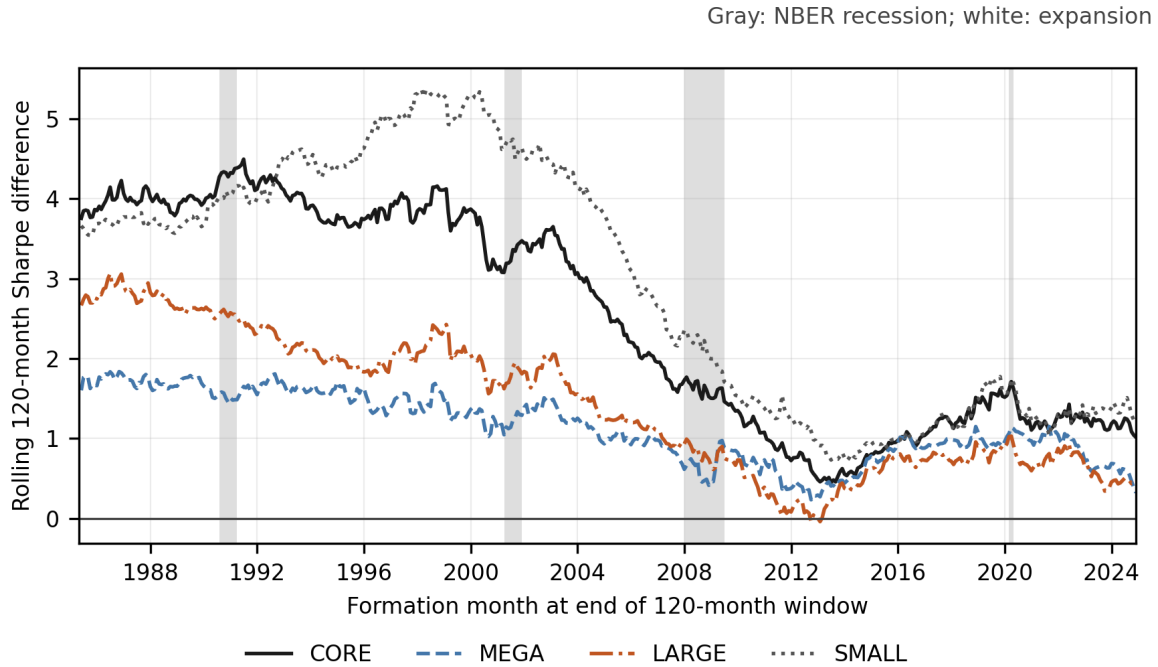


Figure IA.4. Rolling 120-month gross Sharpe-ratio difference, complete IAB minus its intercept-off ablation. The strict-24 sample is June 1975–November 2024, producing 475 overlapping windows. Each point uses the indicated universe’s paired unit-gross return series. The axis labels the last formation month of each window. Gray bands mark NBER recession months; unshaded periods are expansions. Each rolling statistic includes earlier months from potentially different business-cycle phases. Overlapping windows are not independent replications; the plot describes evolution rather than a simultaneous significance test.

The complete and intercept-off rules have a positive rolling Sharpe difference in all 475 archived Core windows. That visual persistence should be read together with the recent-period mean interval in Table VI, which includes zero. A persistent descriptive ranking does not establish a precisely estimated recent expected-return increment.